

Program overview

12-Sep-2023 13:11

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Master Materials Science and Engineering

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Master MSE 20223							
Master Year 1							
MSE Core Programme (obligatory for all students: 41 ECTS)							
CH4011MS	Polymer Science	4					
MS43006	Structure and Properties of Materials	6					
MS43011	Characterisation of Materials	5					
MS43016	Metals Science	3					
MS43020	Computational Materials Science	3					
MS43025	Mechanical Behaviour of Materials	4					
MS43030	Processing of Materials	3					
MS43035	Thermodynamics and Kinetics of Materials	3					
MS43045	Societys Needs: Case Studies and Materials Challenges	4					
WM0320TU	Ethics and Engineering	3					
MSE Specialisation Courses							
MSE Specialisation Materials in Engineering Applications							
MSE MEA Obligatory fixed (6 ECTS)							
MS43100	Science of Failure	3					
MS43120	Corrosion Engineering	3					
MSE MEA Obligatory selection (Select at least 8 ECTS)							
MS43110	Joining Technologies	3					
MS43115	Materials Selection for Engineering Applications	3					
MS43210	Advanced Characterisation	4					
MS43801	Determination of Microstructure	2					
MSE Specialisation Metals Science and Technology							
MSE MST Obligatory fixed (7 ECTS)							
MS43100	Science of Failure	3					
MS43200	Metals Science II	4					
MSE MST Obligatory selection (Select at least 7 ECTS)							
MS43205	Computational Materials Science II	3					
MS43210	Advanced Characterisation	4					
MS43215	Steel Science	3					
MS43220	Corrosion Science	3					
MSE Specialisation Materials for Sustainable Development							
MSE MSD Obligatory fixed (8 ECTS)							
MS43305	Materials for Hydrogen and Solar Energy Technology	4					
MS43315	Recycling Engineering Materials	4					
MSE MSD Obligatory selection (Select at least 6 ECTS)							
MS43205	Computational Materials Science II	3					
MS43310	Materials at High Temperature	4					
MS43330	Sustainable Metals Production	4					
MSE Free Specialisation 14 EC (to be filled in consultation with master coordinator)							
MSE Recommended electives							
AE4ASM001	Design of lightweight structures I: Composites & Metals	3					
AE4ASM004	Manufacturing of Aerospace Structures & Materials	3					
AE4ASM005	Fatigue of Structures & Materials	3					
AE4ASM103	Functional Coatings	3					
AE4ASM109	Design & Analysis of Composite Structures I	5					
AE4ASM504	Structural Integrity and Maintenance	3					
AE4ASM508	Design of Self-healing materials	3					
AE4ASM510	Design & Analysis of Composite Structures II	3					
AE4ASM516	Material Selection in Mechanical Design	3					
BM41035	Biomaterials	4					
CH3783	Materials Chemistry for The Nuclear Fuel Cycle	3					
ET4376	Photovoltaic Basics	4					
ET4377	Photovoltaic Technologies	4					
ID5356	Sustainable Design Strategies for Product Development	3					
MS43040	Machine Learning for Materials Design	4					
MS43100	Science of Failure	3					
MS43110	Joining Technologies	3					
MS43115	Materials Selection for Engineering Applications	3					
MS43120	Corrosion Engineering	3					

MS43200	Metals Science II	4
MS43205	Computational Materials Science II	3
MS43210	Advanced Characterisation	4
MS43215	Steel Science	3
MS43220	Corrosion Science	3
MS43305	Materials for Hydrogen and Solar Energy Technology	4
MS43310	Materials at High Temperature	4
MS43315	Recycling Engineering Materials	4
MS43325	Application of Materials in High Tech Engineering	3
MS43330	Sustainable Metals Production	4
MS43801	Determination of Microstructure	2
MS43805	Material and Fabrication Factors in Marine Structures	5
MS43810	Materials in Art and Design	3
MS43825	Radiation Damage in Materials	4
SET3055	Economics and Regulations of Sustainable Energy Systems	4

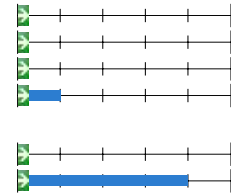
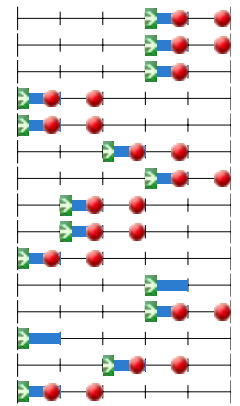
Master Year 2

MSE MSc Projects and Assignments

MS43702	Materials Elective Project I	2
MS43706	Materials Elective Project II	6
MS53200	External Research Project	15
TUD4040	Joint Interdisciplinary Project	15

MSE MSc Graduation Project (45 ECTS)

MS53010-20	MSE Literature Research	10
MS53035	MSE MSc Thesis	35



Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Master Materials Science and Engineering

Master MSE 20223	
Program Title	Master Materials Science and Engineering
Program Coordinator	The coordinator is the person for questions or problems related to the individual study programme and for monitoring progress. The coordinator can be contacted through the email: MSEcoordinator-3mE@tudelft.nl Dr. Yaiza Gonzalez-Garcia (Program Coordinator and Director of Education) Saskia van der Meer (Education Assistant)
Prerequisites	There are several ways to be admitted to the MSc program Materials Science and Engineering: - As a continuation of an academic BSc program, either in the Netherlands or abroad. - After completing a bachelor's program at a Dutch polytechnic institute or the Netherlands Defence Academy (NLDA). More details about the admission and procedures are found in: https://www.tudelft.nl/onderwijs/opleidingen/masters/mse/msc-materials-science-engineering/admission-and-application
Introduction 1	Materials Science & Engineering (MSE) is an interdisciplinary field focusing on the relationship between structures and the behaviour of materials (science) and the sustainable design of these structures to produce predefined properties for specific applications (engineering). Students will consider materials over a wide range of length and time scales from the behaviour of collections of individual atoms to macroscopic structural and functional properties and from raw material processing through sustainable production and application of materials, to failure, durability and recycling. Students will learn to understand the behaviour of materials under different conditions and assess their suitability in products and industrial processes. Moreover, students will study how to design material properties at nano and micro levels in a suitable manner to suit applications on different scales. The two-year program (120 EC) is taught in English and incorporates a combination of academic learning and active research. Course modules are delivered by lectures, lab sessions, workshops and seminars.
Program Goals	The aim of the program is to offer students high-quality, multi-disciplinary education, and to turn out graduates capable of making immediate and significant contributions to a wide range of industrial and academic areas at a global level. The main goals are: 1. to provide students with a sound and thorough understanding of the underlying scientific and engineering principles involved in Materials Science and Engineering 2. to enhance knowledge of sustainable materials design, selection, processing and characterisation 3. to build up awareness of the environmental, economic and human aspects of materials selection, usage, recycling and disposal.
Exit Qualifications	A graduate in Materials Science & Engineering meets the following qualifications: 1. Profound knowledge of Materials Science and Engineering and the underlying engineering sciences (mathematics, physics and chemistry) 2. The capability to apply this advanced knowledge effectively to reach the forefront of scientific or industrial research and development. 3. The skills to actively and independently apply this knowledge for analysing, modelling, simulating, designing and performing research with respect to problems related to Materials Science and Engineering 4. The capability to solve technological problems in a systematic way. Professional attitude towards identifying and acquiring lacking expertise, and monitoring and critically evaluating existing knowledge. 5. The capability to work both independently and in multidisciplinary teams. Effective communication with both professionals and a non-specialised public. 6. The capability to evaluate and assess the technological, ethical and societal impact of ones work, and to take responsibility regarding sustainability, economy and social welfare.
Program Structure 1	The program is structured as followed: - Core Program (41 EC) obligatory set of courses which provides the basic knowledge required by all graduates in Materials Science and - Specialisation (34 EC) focusing on a selected area of interest - Literature Research (10 EC) - Master Thesis Project (35 EC): independent scientific investigation that leads to a thesis. The specialisation focuses on science, engineering and application-related issues based on the interest of the student. The predefined specialisations are: 1. Materials in Engineering Applications (MEA) 2. Metals Science and Technology (MST) 3. Materials for Sustainable Development (MSD) The student can also define their own specialisation upon approval by the program coordinator and the Board of Examiners.

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Master Year 1

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MSE Core Programme (obligatory for all students: 41 ECTS)	
Introduction 1	Core Programme The Core Programme, that runs through the first semester and a small part of the second semester, contains identical modules for all students. It focuses on Materials Science and Engineering fundamentals and general knowledge. It offers students a firm grounding upon which they can build more specialised or wider ranging interests. This core programme comprises the compulsory modules shown below (total of 41 EC).

CH4011MS	Polymer Science	4
Responsible Instructor	Prof.dr. S.J. Picken	
Responsible Instructor	Dr. E. Mendes	
Instructor	Dr. G.A. Filonenko	
Contact Hours / Week x/x/x/x	The course description will be available soon.	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Course Contents In this course, you will be taught the basics of polymer science starting from definitions and various kinds of molecular structures of polymers until their properties in solution, melt and bulk material such as in elastomers, gels and polymer glasses. You will be taught how universal polymer chain models (ideal chain and excluded volume chain) can describe molecular behavior and chain conformations in bulk or solution for polymers with any chemical structure. Visco-elasticity of polymer solutions in dilute, semi dilute and melt will be considered from non-entangled to entangled regime in solutions and melts. Chain dynamics, reptation theory and tube models will be considered and you will be taught how to interpret experimental data in the light of those theories. You will learn how to extend visco-elasticity predictions to the glassy state and how free volume theories describe a polymer glass. Polymer miscibility and polymer blends, block copolymers self-assembly in solution and in melt and other ordered phenomena in polymers (crystallinity and liquid crystallinity) will also be taught. Finally, main aspects of polymer composites will be taught. We will give some examples of how polymers are applied and which one(s) to choose for the purpose.	
Study Goals	Study Goals Knowledge 1) be able to explain universal polymer chain models and chain statistics in relation to chemical structure 2) be able to explain what are the main quantities related to polymers such as, persistence length, radius of gyration, reptation time, glass transition temperature, etc. 3) be able to describe how various polymer characterisation methods work Understanding 4) be able to explain the role of entropy and of excluded volume interactions in polymer conformations and visco-elasticity 5) be able to formulate the concepts of miscibility, solvent quality and phase separation in polymer solutions and polymer blends 6) be able to formulate the concepts of entanglements and polymer tube Analysis (critical thinking) 7) be able to interpret quantitatively (characterization) experimental data on polymer materials 8) be able to calculate basic quantities related to structure and dynamics of polymer materials Application 9) be able to select or propose a polymer for a given application based on knowledge acquired in the course. 10) have a general overview of what polymers are used for, also discussing some examples.	
Education Method	Lectures	
Literature and Study Materials	Book part 1: Physics of Polymers M. Rubinstein & R. Colby Oxford Book part 2: From Polymers to Plastics A.K. van der Vegt VSSD 2002 Delft University Press	
Assessment	written exam with open slides	
Elective	Yes	

MS43006	Structure and Properties of Materials	6
Responsible Instructor	Dr. M.H.F. Sluiter	
Instructor	Dr. M.W.E.M. Alfeld	
Instructor	Dr. P. Dey	
Co-responsible for assignments	Dr. P. Dey	
Co-responsible for assignments	Dr. M.W.E.M. Alfeld	
Contact Hours / Week x/x/x/x	2.4.0.0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Required for	most other courses in the materials science curriculum	
Expected prior knowledge	THE LIST BELOW IS VERY IMPORTANT! IT IS THE STUDENTS' RESPONSIBILITY TO START WITH PROPER PRIOR KNOWLEDGE. vectors & matrices (basics of linear algebra) complex numbers (square root of -1 etc. etc.) calculus Fourier transform two pdf texts with details and a self-test are available under "content" on Brightspace: MatSci - preknowledge mathematics MatSci - entry level test mathematics high school level chemistry, e.g. (but not limited to) -basic understanding of the periodic table of the elements -concept of valence -chemical bond high school level physics, e.g. (but not limited to) -concepts of energy, potential energy, kinetic energy -Newton's laws -electrostatics -law of gravitation -laws of thermodynamics	
Course Contents	part 1 (1st quarter taught by Marcel Sluiter) Crystals and Crystallography (chapter 4 Naumann textbook) Crystals (primitive translations, unit cell, lattice points, basis, Miller (Bravais) indices, interplanar spacing) Crystal Systems and Symmetry Structural Relationships (density and packing, atomic density, mass density, ...) Interstices Quasicrystals The Structure of Matter (chapter 5 Naumann textbook) metals Intermetallic Compounds Ionic Compounds Covalent Structures Structure of Glass Structure of Polymers Quantum Mechanics (chapter 2 Naumann textbook) Atomic Structure The Electron (de Broglie, Heisenberg uncertainty) Schrödinger Wave Equation One Electron Approximation Periodic Table Chemical Bonding (chapter 3 Naumann textbook) Ionic Bonding Covalent Bond Metallic Bond Atomic and Ionic Radii Secondary Bonding Other Potential Functions ----- mid term exam over material covered in part 1 ----- Part 2 (2nd quarter taught by Poulumi Dey and Matthias Alfeld) Physics and Chemistry of Materials Ch15 -Ch22 Naumann covering: Distribution Functions Lattice Vibrations and Phonons Thermal Properties of Solids Free Electrons in Metals Band Theory of Metals Semiconductors Theory and Application of Junctions Transistors, Quantum Wells and Superlattices	
Study Goals	The student is able to rationalize materials properties and relate them to atomic and electronic structures.	

Education Method	Q1 part: lectures and practice sessions Literature and Study Materials Introduction to the Physics and Chemistry of Materials, by Robert J. Naumann, CRC Press, 2008, ISBN-13:9781420061338, ISBN-10:142006133X (Chapters 1-5) additionally some other course material will be offered in the form of pdf files during the course. Q2 part: lectures and practice sessions. Literature and Study Materials Introduction to the Physics and Chemistry of Materials, by Robert J. Naumann, CRC Press, 2008, ISBN-13:9781420061338, ISBN-10:142006133X (Chapters 15-22)
Computer Use	microsoft office (word, powerpoint, excel) or equivalent simple text editor
Literature and Study Materials	Introduction to the Physics and Chemistry of Materials, by Robert J. Naumann, CRC Press, 2008, ISBN-13:9781420061338, ISBN-10:142006133X additionally some other course material will be offered in the form of pdf files during the course.
Books	Introduction to the Physics and Chemistry of Materials, by Robert J. Naumann, CRC Press, 2008, ISBN-13:9781420061338, ISBN-10:142006133X
Assessment	Written exams, on-campus (possibly shifted to online, depending on the Covid safety situation in Q1 part 1 and Q2 part 2). Concerning Part 1: Closed-book multiple choice questions and numerical questions. Concerning Part 2: On-campus exams will be closed book with a focus on multiple choice questions. Online exams will be open book with open questions. Resits in Q2 (part 1) and Q3 (part 2) under identical conditions.
Permitted Materials during Tests	generally closed-book exams
Tags	Algebra Analysis Calculus Chemistry Energy Physics Quantum
Contact	part 1: Marcel Sluiter: M.H.F.Sluiter@tudelft.nl part 2: Matthias Alfeld: M.Alfeld@tudelft.nl

MS43011	Characterisation of Materials	5
Responsible Instructor	M.J. Santofimia Navarro	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	In the Characterisation of Materials course, general research tools in materials science are introduced. The following modules constitute the course, although the number and content of modules may be slightly modified upon starting of the course. If that is the case, students will be informed through Brightspace. - Module 1: Optical Microscopy - Module 2: Electron Microscopy - Module 3: Thermal analysis - Module 4: Mechanical Behaviour - Module 5: Electrochemistry - Module 6: X-ray diffraction	
Study Goals	On completion of the course, the student will be able to select techniques for materials characterisation, describe those techniques, specify the physical principles involved and interpret the results obtained. More specifically, the students will be able to: - use optical microscopy for materials characterisation. This includes sample preparation and selection of appropriate etching agents, - describe the principles and justify the use of scanning electron microscopy and energy dispersive x-ray spectroscopy in materials research and analyse the data, - derive information on phase transformations from dilatometry experiments, compare the results to CCT diagrams and correlate the data with the observed microstructure and the phenomena occurring during welding and heat treatments, - select appropriate mechanical tests and analyse the data, - identify corrosion mechanisms and quantify parameters which influence the corrosion rate, - describe the principles of the x-ray diffraction technique and analyse the data, - communicate in writing the experimental condition of the specific techniques and results obtained. Write different type of reports as requested, per lab class. Small modifications in the content and number of modules may lead to a slight modification and/or reduction of the goals above. In that case, students will be informed via Brightspace.	
Education Method	Lectures and lab sessions.	
Literature and Study Materials	Literature, readers and study materials will be provided in the course.	
Assessment	The assessment of the course includes: - Group assignments (70% of the grade). - An individual open-book multiple-choice exam (30% of the grade). Students are required to obtain a minimum mark of 6 for both parts in order to pass the course. Peer-reviewing will be considered in the final grade (+/- 1 point) The online course includes individual questionnaires and quizzes for self-assessment that will not contribute to the grade.	
Permitted Materials during Tests	During the open-book multiple choice examination, it is permitted to use any material from the course. The student can use laptop or tablet to access course material. However, the access to internet and the use of telephone is not allowed. It is strongly advised that the student brings electronic devices fully charged to the exam, since in some classrooms the access to electric sockets may be limited.	
Department	3mE Department Materials Science & Engineering	

MS43016	Metals Science	3
Responsible Instructor	Prof.dr.ir. L.A.I. Kestens	
Instructor	Ir. N. Geerlofs	
Contact Hours / Week x/x/x/x	0.4.0.0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Metals represent a vital class of materials for a technological society. This course examines the structure and properties of metals across a range of length scales, addressing issues of microstructural changes and phase transformations, metals production techniques and the behaviour of metals in generic applications. The course covers microstructures, mechanical properties in relation to microstructures and crystallographic aspects of physical metallurgy. Microstructural aspects include: 1) basic elements of the plasticity of metal alloys including concepts such as the theoretical strength of single crystals, plasticity by dislocation motion in single crystals, stress fields induced by dislocations, dislocation energy, force on dislocations (Peach-Koehler law), multiplication of dislocation (Frank-Read source). 2) Strengthening mechanisms: work hardening (Taylor equation), grain-boundary strengthening (Hall-Petch relation), solid solution strengthening, precipitation hardening, age-hardening. 3) Various stages of work hardening. Dynamic recovery, effect of stacking-fault energy. Shockley partial dislocations. 4) Basics of crystal plasticity theory: Taylor vs Sachs theory (introduction). 5) The crystallographic theory of twinning. 6) the phenomenological theory of martensitic transformations, occurring under either thermal or mechanical driving force. 7) the origin of crystallographic texture in metallic microstructures, the representation of texture and the experimental techniques to measure texture on a macro- or microscopic scale. The typical textures of steel and aluminium alloys (rolling, annealing and transformation textures). 8) The deformation microstructure: cell formation, deformation bands, shear bands, stored energy of plastic deformation 9) Thermally activated recovery mechanisms: recovery (formation of low-angle grain boundaries), recrystallization (nucleation and growth), JMAK equation, curvature driven grain growth (Gibbs-Thompson relation), Zener pinning.	
Study Goals	1. Understand the basic deformation mechanisms of materials. 2. Acquiring knowledge about the mechanical behaviour of materials in relation to the fundamental deformation mechanisms. 3. To designate elements in the material design that affect strengthening behaviour. 4. Acquiring knowledge concerning the crystallographic structure and properties of mechanical twins. 5. To gain insight in the process of martensitic phase transformation. 6. Being able to establish a link between the typical characteristics of the (sub-)structure of a plastically deformed metal and the properties of the recrystallised structure after annealing. 7. Mastering the basic concepts of quantitative texture analysis and being able to identify and analyse the crystallographic texture of standard steel and aluminium grades.	
Education Method	Standard white board lectures (either on-line or on-campus, pending on faculty regulation) backed-up by slides or textbooks. Q&A sessions to apply the lectures in problem solving exercises.	
Assessment	Standard: on-campus exam (closed book) with open questions or application oriented questions; If exam is online (because of faculty regulation): open book examination with application oriented questions only. Typical duration of exam: 3h-3.5h	

MS43020	Computational Materials Science	3
Responsible Instructor	Dr. M.H.F. Sluiter	
Co-responsible for assignments	Dr.ir. C. Bos	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	MS43205 Computational Materials Science 2.	
Expected prior knowledge	Basic knowledge of algorithms, basics of computer programming, a working knowledge of PYTHON, basics of Newtonian mechanics (3 laws), quantum mechanics (wave function, Schroedinger eqn, uncertainty principle); basics of thermodynamics; calculus. Also some familiarity with basic materials science concepts such as crystal structures - translations & periodicity, crystal basis, vacancy, dislocation, grain boundary, grain coarsening, phase transformation such as solidification, spinodal decomposition, nucleation. If you have no prior experience with Python you can follow a self-study course at https://www.learnpython.org/ (of course, you may also use other Python courses/resources to prepare yourself.) Take note: * MS43020 requires that you can use Python to write simple computer programs BEFORE you start the course. * Before you start the course you must verify your Python skills by taking a test (location to be specified) Make this test in Q1, so that you may remedy your lack of skills BEFORE Q3. NB It is NOT recommended to skip this course in the 1st year of the MSc because then you will have two chances at passing the exam only. Much better is to prepare in a timely fashion and take part in the course from the start.	
Course Contents	Parts 1 and 2 of the course text: Computational Materials Science: Fundamentals to Applications by R. LeSar (ISBN978-0-521-84587-8). Home work and many of the illustrations utilize Jupyter notebooks. Modeling and simulation What is meant by computational materials science and engineering? Scales in materials structure and behavior How to develop models The random-walk model Random-walk model of diffusion Connection to the diffusion coefficient Bulk diffusion A random-walk simulation Random-walk models for materials Simulation of finite systems Sums of interacting pairs of objects Perfect crystals Cutoffs Periodic boundary conditions Implementation Long-ranged potentials Electronic structure methods Quantum mechanics of multielectron systems Early density functional theories The Hohenberg-Kohn theorem Kohn-Sham method The exchange-correlation functional Wave functions Pseudopotentials Use of density functional theory Interatomic potentials The cohesive energy Interatomic potentials Pair potentials Ionic materials Metals Covalent solids Systems with mixed bonding What we can simulate Determining parameters in potentials Molecular dynamics Basics of molecular dynamics for atomic systems An example calculation Velocity rescaling Molecular dynamics in other ensembles Accelerated dynamics Limitations of molecular dynamics Molecular dynamics in materials research The Monte Carlo method Ensemble averages The Metropolis algorithm The Ising model Monte Carlo for atomic systems Other ensembles Time in a Monte Carlo simulation Assessment of the Monte Carlo method	

	Uses of the Monte Carlo method in materials research
Course Contents Continuation	computational materials science II (MS43205) is a follow-up course. In the follow-up the 2nd part of the textbook by LeSar is used.
Study Goals	After passing the course, the student will be able to... 0) can write a piece of PYTHON code that represents a model of materials behavior 1) determine what type of computational model is needed to simulate a particular phenomenon in materials science 2) understand the basic principles behind several classes of models, such as stochastic models involving random-walk, and Monte Carlo sampling simulations involving finite systems, e.g. supercell modelling as used in many ab initio electronic structure methods and molecular dynamics 3) understand, apply, interpret and analyze results from a selection of models: diffusion through random walk interatomic interactions in a lattice gas through supercell calculations using a electronic density functional method also called "ab initio method" Monte Carlo sampling of the partition function force fields as a glue between ab initio and empirical information on the one side and dynamics over extended time and length scales on the other side Molecular dynamics simulations involving force fields 4) visualize data for insight and for communication
Education Method	Lectures & homework computer exercises (TU Delft computers can be used)
Computer Use	Yes, for exercises you can use TU Delft computers.
Books	Computational Materials Science: Fundamentals to Applications by R. LeSar (ISBN978-0-521-84587-8).
Prerequisites	* MS43020 requires that you can use Python to write simple computer programs BEFORE you start the course. * Before you start the course you must verify your Python skills by taking a test (location to be specified) Make this test in Q1, so that you may remedy your lack of skills BEFORE Q3. If you have no prior experience with Python you can follow a self-study course at https://www.learnpython.org/ (of course, you may also use other Python courses/resources to prepare yourself.)
Assessment	Computer-based exam. The exam consists of exercises where you write Python code to simulate phenomena in materials - similar in style and format to the homework exercises during the course.
Permitted Materials during Tests	TUD PC.
Tags	Algorithmics Analysis Artificial intelligence Calculus Computer Engineering Modelling Numeric Methods Programming Programming concepts Video
Department	3mE Department Materials Science & Engineering

MS43025	Mechanical Behaviour of Materials	4
Responsible Instructor	Dr. S. Kumar	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course provides the essential knowledge required to understand and predict the mechanical behaviour of materials. The following aspects are addressed and illustrated with reference to a specific material class (in brackets): Tensors and kinematics Generalized elasticity Viscoelasticity (polymers) Plasticity (metals) Finite-strain hyperelasticity (rubber/biological tissues)	
Study Goals	Upon completion of the course, students should be able to: 1. Model elastic behavior of isotropic and anisotropic materials 2. Model time dependent elastic behavior of polymers 3. Model metal plasticity using yield surfaces such as Tresca and Von Mises 4. Model behavior of rubber and biological tissues under large strains	
Education Method	Lectures and exercises	
Literature and Study Materials	Lecture notes will be provided on Brightspace. Additional reading material: "The Mechanics of Constitutive Modeling" by Niels Ottosen, Matti Ristinmaa Note that above reading is optional/supplemental and not a textbook for this course.	
Assessment	Two homework assignments will be given at the mid-point and end of the course. Each assignment will account for 10% of the final grade (cumulatively, 20%). The rest 80% of the final grade will be assessed by a written final examination. Note: Exam will be conducted with open lecture-notes. Books and electronic devices are not allowed. Students are advised (not mandatory) to make their own concise formula sheets for the exam. Simple calculators are allowed. Graphing calculators are not allowed.	
Tags	Mechanics Modelling	
Department	3mE Department Materials Science & Engineering	
Contact	Contact by email: Sid.Kumar@tudelft.nl	

MS43030	Processing of Materials	3
Responsible Instructor	Dr. N.D. Dogan	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Subjects covered include transport phenomena in materials processing. The course focuses on quantitative descriptions of fluid flow and heat transfer applied to different materials processing steps and applications. The following topics will be discussed during the lectures: Module I: Fluid dynamics Viscous properties of fluid Laminar flow and the momentum equation Turbulent flow and complex flows Energy balance applications in fluid flow Module II: Energy transport Fouriers law and thermal conductivity Heat transfer and the energy equation Correlations and data for heat transfer coefficient Conduction of heat in solids Solidification of metals	
Study Goals	By the end of this course, you should be able to: - Explain basic principles of heat and energy transport - Formulate materials processing problems in terms of mathematical equations - Solve simplified problems in materials processing	
Education Method	The course activities consist of: On campus lecture, including exercises & homework assignments (reading and problem solving).	
Assessment	In order to successfully complete the course, you will need to score minimum 6 out of 10 points in the final grade. This final grade consists of two homework assignments. these will be given at the end of each module & together account for 20% of the total grade. The rest 80% will be assessed by a written examination, open book.	
Department	3mE Department Materials Science & Engineering	

MS43035	Thermodynamics and Kinetics of Materials	3
Responsible Instructor	M.J. Santofimia Navarro	
Contact Hours / Week x/x/x/x	4.0.0.0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course introduces the thermodynamics and kinetics of phase transformations. Thermodynamical subjects include Gibbs free energy, chemical potential, ideal and real solutions, binary phase diagrams, Gibbs phase rule, Gibbs-Thomson effect, ternary systems, and ternary phase diagrams. Kinetics refer to atomic mechanism of diffusion, substitutional and interstitial diffusion, high-diffusivity path, and diffusion in ternary alloys. The different type of interfaces that can be found in a microstructure are also treated in the course (free surfaces, grain boundaries and phase interfaces). The course also introduce mechanisms associated to phase transformations in the solid state, such as nucleation and growth, as well as basic concepts on phase transformations in steels.	
Study Goals	A the end of this course students will be able to: 1. Formulate and apply the thermodynamics of unary systems and of binary systems based on the ideal and regular solution models. 2. Interpret equilibrium phase diagrams and formulate the thermodynamics behind them in terms of Gibbs free energy, enthalpy, entropy and chemical potential. 3. Describe and identify the process of atomic diffusion in crystalline structures. 4. Describe and identify different types of interfaces and estimate their interfacial energies in crystal structures. 5. Relate thermodynamics, diffusional processes, interface types and presence of secondary particles with the occurrence and progress of grain growth and solid-solid phase transformations. 6. Formulate solid-solid phase transformations in terms of nucleation and growth and describe their progress based on TTT diagrams and Avramis model. 7. Describe basic characteristics of solid-solid phase transformations in steels.	
Education Method	Lectures and applied questions.	
Assessment	Close book written exam.	

MS43045	Societys Needs: Case Studies and Materials Challenges	4
Responsible Instructor	P. Taheri	
Instructor	Dr. G.H.J. Langejans	
Instructor	S.T. Abrahami	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	This course focusses on the challenges related to realising truly sustainable usage of materials in technology in terms of resources, energy, waste and society/stakeholders. Each group of four students will choose a grand challenge in materials sustainability. Students will analyse the life cycle of involved material(s). In doing so, the application, product requirements and material properties are quantitatively analysed. Students will consider aspects like the used resources and energy costs in producing the properties needed for the required function, (long-term) performance, as well as the waste emission, including CO2, during the entire life cycle of production, use and recycling. Students will research and question what sustainability and decolonial sustainability is, and they will reflect on the impact of current and alternative materials for a range of stakeholders. Students will study the microstructure of the material in their grand challenge The study is to be concluded by discussion on current limitations and future directions. The aim of this course is threefold. One, to gain insight into the physics and technological aspects of the design and application of a material. Two, to develop consciousness of and insight into the role of materials in sustainable technology and societal use of materials. Three, develop consciousness of own biases and sensitivities towards other views. Lectures plan Session 1: Introduction to the course and project topics Session 2: Materials sustainability and introduction to Life Cycle Assessment (LCA) Session 3: Stakeholder analysis Session 4: Feedback workshop Session 5: Students mid term presentation Session 6: LCA case-studies practice Session 7: Historic and colonial perspective on materials and sustainability Session 8: Personal reflection on historic and colonial perspective in individual projects.	
Study Goals	By the end of the course the students are able to: - analyse and quantify sustainability aspects of the production, the usage and the recycling of materials in a specific application (i.e. grand challenge), - design and formulate concepts for to improve sustainability, - Are aware of the bigger context of global challenges in terms of environment, unseen stakeholders, gender, economy and culture, - Reflect on historical (colonial) past & legacy and its relationship with the project, - critically read and assess scientific literature and consult experts on the topic, - present and defend scientific materials research based on a literature search in the form of an oral presentation and in the form of a written report, - work in groups on a materials oriented project, student is able to give and receive constructive feedback from fellow students.	
Education Method	Compulsory weekly lectures are held in parallel to self-organised group research and compulsory update meetings with the lecturers. Groups of students are composed by the course lecturer. Each group works on a grand challenge, to be chosen by the students. The students obtain and research relevant information from textbooks, scientific papers, and discussions with experts.	
Assessment	The assessment is based on: - Participation in the weekly lectures/workshops (20%) - Written report on the analysed grand challenge, LCA and decolonial reflection (40%) - Final presentation on the analysed grand challenge, LCA and decolonial reflection (40%) The students also present their work during a mid-term symposium. The mid-term symposium is to show progress and is an opportunity to learn from other groups and engage in discussion with other students and lecturers, whereas the final symposium will be graded	
Department	3mE Department Materials Science & Engineering	

WM0320TU	Ethics and Engineering	3
Module Manager	Dr. F. Santoni De Sio	
Instructor	M. Sand	
Co-responsible for assignments	M. Sand	
Contact Hours / Week x/x/x/x	4/0/4/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Course Contents	This course is identical to the initial part of the course WM0329TU. You will explore the ethical and social aspects and problems related to technology and to your future work as professional or manager in the design, development, management or control of technology. You will be introduced to and make exercises with a range of relevant aspects and concepts, including professional codes, philosophical ethics, individual and collective responsibilities, value pluralism and relativism, responsibility within organisations, responsible conduct of companies and the role of law. You will analyse legal, political and organisational backgrounds to existing and emerging ethical and social problems of technology, and you will explore possibilities for resolving, diminishing or preventing these problems.	
Study Goals	After having completed the course you: can better recognise and analyse ethical and social aspects and problems inherent in technology and in the work of professionals and managers active in the design, development, management and control of technology. have insight into how these ethical and social aspects and problems are related to legal, political and organisational backgrounds. are able to explore and assess possibilities for solving or diminishing existing and emerging ethical and social problems that attach to technology and the work of professionals and managers. are better prepared to perform your future work as a professional or manager in the design, development, production and control of technology in an ethical and socially responsible way.	
Education Method	A series of 6 lectures and interactive work sessions with partial assessment, concluded with a written test.	
Literature and Study Materials	Chapters from the Reader Ethics and Engineering, available at Nextprint and as PDF files on Brightspace; articles available as PDF on Brightspace; Powerpoint slides, materials used in the working groups, and lecture notes.	
Assessment	Written exam (60%), presentation and active participation during the working group sessions (40%). In order to pass the course, a minimal grade of 5.5 should be obtained *both* in the exam *and* in the working group, and the average grade should be higher than 5.75. It is possible to re-take the exam without re-taking the working groups, in this case the working group grade will be kept. If you re-take the exam, it is also possible to re-attend the working groups to improve the final grade.	
Enrolment / Application	Enrolment via Brightspace is required for this course. This is needed in order to plan the workgroups. Please enroll as soon as possible and check Brightspace for more information on the enrolment on the enrolment procedure and timeline. The Brightspace page will be activated a couple of weeks before the start of the course.	
Remarks	The course is run twice each year in the first and third quarter. The course is identical to the initial part of the course wm0329tu (6 ects).	
Elective	Yes	
Category	MSc niveau	

MSE Specialisation Courses

Introduction 1

Specialisation Course

In the second semester, students start to take a Specialisation Course based on their own interests. This course focuses on science, engineering and application-related issues within a specific theme and has a total size of 34 EC. At the end of the first semester students choose their Specialisation Course.

Pre-defined Specialisation Courses

A number of Specialisation Courses have been pre-defined. Each pre-defined Specialisation Course comprises obligatory modules, possibly some modules of which at least a certain number should be chosen, and elective modules (see below).

Elective Modules

An overview of recommended elective modules for each of the pre-defined Specialisation Courses can be found in the respective descriptions. Taking these elective modules will per definition be approved by the Board of Examiners. Other elective modules, however, require explicit approval in order to assure that the acquired knowledge and skills contribute to the exit qualifications of the MSE master's programme. Students should take notice of the following:

- Any of the following items can potentially be regarded as an elective module, provided its level and weight are sufficiently high:
 - a module scheduled at Delft University of Technology;
 - a special assignment or project defined and supervised by a scientific staff member of the Department MSE or someone participating as lecturer in the master's programme MSE;
 - an external research project, organised by an external organisation or institution, supervised by local staff, and approved and assessed by a scientific staff member of the MSE department or someone participating as lecturer in the MSE master's programme.
- Before taking elective modules, each student should present his or her programme to the Board of Examiners for approval. Changes can be allowed at a later stage, but only after renewed approval by the Board of Examiners.
- Cooperation between students during a minor part of an assignment or project is allowed, but each elective module is assessed on an individual basis. This means that each assignment or project should be a unique piece of work and lead to an individual report (and presentation).

Self-defined Specialisation Courses

Students can also choose to define their own Specialisation Course. Examples of themes for self-defined Specialisation Courses are composites, biomaterials, polymers, functional ceramics and materials & nanotechnology. Modules for these self-defined Specialisation Courses can be offered by the MSE department or by other departments at TU Delft.

Students should get approval for self-defined Specialisation Courses by the Board of Examiners. To get this approval students should take notice of the following:

- The theme of the Specialisation Course should not be covered by one of the pre-defined Specialisation Courses;
- At least 14 EC of the modules should form a consistent set related to the theme of the Specialisation Course;
- The majority of the modules should connect to the exit qualifications of the master Materials Science and Engineering;
- Students should write a letter in which they motivate why they want to define their own Specialisation Course and how they will deal with a possible lack of knowledge.

Notes:

- Student activities that are mainly undertaken to acquire work experience or to make a first acquaintance with job conditions cannot be regarded as elective modules.
- The supervisor of the Master Thesis Project has no role in approving or disapproving a student's proposed Specialisation Course. This is simply because students often start their Specialisation Course without already having chosen a thesis supervisor. The thesis supervisor may, however, require passing certain modules as part of the thesis research project.

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Master Materials Science and Engineering

MSE Specialisation Materials in Engineering Applications

Introduction 1

Materials in Engineering Applications (MEA)

The automotive, offshore, maritime, medical and various other industries all depend on advanced materials technologies. Indeed, innovations in these fields often require materials that must meet contradictory, nearly impossible demands. In this Specialisation Course you will learn how to optimise materials for engineering applications by considering properties with respect to aspects such as production, joining, corrosion and failure. Alternatives for materials processing and the effect of this processing on properties will be addressed.

This Specialisation Course comprises the following obligatory modules:

-) Science of Failure (MS43100)
-) Corrosion Engineering (MS43120)

Furthermore at least three of the following modules should be taken:

-) Determination of Microstructures (MS43801)(You will also need Advanced Characterisation for this course)
-) Joining Technologies (MS43110)
-) Advanced Characterisation (MS43210)
-) Materials Selection for Engineering Applications (MS43115)

Finally, up to 20 EC of elective modules should be chosen.

Below details of all modules, including the recommended elective modules, of this Specialisation Course can be found.

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Master Materials Science and Engineering

MSE MEA Obligatory fixed (6 ECTS)
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MS43100	Science of Failure	3
Responsible Instructor	V. Popovich	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	MS43025 - Mechanical Behaviour of Materials or equivalent knowledge	
Summary	Knowledge, tools and procedures to describe and assess fracture mechanical behaviour of materials.	
Course Contents	Roughly, the outline is as follows (# of lecture hours between brackets): Linear Elastic Fracture Mechanics: stress intensity factor, crack-tip plasticity, energy release rate (4) Elastic Plastic Fracture Mechanics: crack tip opening displacement, J integral (4) Failure: ductile and brittle fracture, ductile-to-brittle transition curve, failure assessment diagram (6) Fatigue: crack growth and crack initiation (4) Environment-assisted cracking: mechanisms and testing (2) Instructions: exercises in class (6) This totals 26 hours.	
Study Goals	The student is able to employ basic fracture mechanical quantities to describe crack growth in solids. Furthermore, the student is able to identify a number of common mechanical phenomena that cause material failure in terms of the mechanisms and the conditions for which such behaviour can be expected. The student understands how to experimentally determine relevant material properties and can make simple failure predictions. More specifically, the student is able to: 1. identify the principles, limitations and applications of the fracture mechanical concepts stress intensity, energy release rate, crack tip opening displacement and J integral; 2. explain experimental procedures to quantify fracture mechanics parameters; 3. identify the major microstructural aspects of brittle and ductile fracture in metals; 4. describe failure assessment in case of ductile and brittle fracture; 5. describe fatigue crack growth with delta K, explain the concept of crack closure, predict fatigue lifetime for constant amplitude loading and illustrate the effect of variable amplitude loading; 6. describe the mechanisms of fatigue crack initiation and relate this to surface roughness and microstructure; 7. illustrate the mechanisms for environment-assisted cracking in metals; 8. explain the principles for quantifying environmentally assisted cracking behaviour and predict lifetime under constant load.	
Education Method	Lectures and instructions	
Literature and Study Materials	Fracture Mechanics, M. Janssen, J. Zuidema and R.J.H. Wanhill, 2nd edition, DUP (2002) Collection of Exercises on Fracture Mechanics, available on Brightspace Slides (including notes) on Brightspace	
Assessment	Written examination during which the book Fracture Mechanics and the hand-outs provided may be consulted. Other forms of information, such as worked-out problems, are not allowed!	
Tags	Industry Mechanics Physics Research Methods Structural Mechanics Sustainability Technology	
Department	3mE Department Materials Science & Engineering	

MS43120	Corrosion Engineering	3
Responsible Instructor	Prof.dr.ir. J.M.C. Mol	
Co-responsible for assignments	P. Taheri	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Summary	corrosion, corrosion principles, corrosion types, corrosion prevention and protection	
Course Contents	- Relevance of corrosion - Costs to society - Definitions and electrochemical character of corrosion - General corrosion vs local forms of corrosion - Electrochemical Thermodynamics - Electrochemical Kinetics - Passivity - Galvanic Corrosion and intergranular corrosion - Pitting and Crevice Corrosion - Protection against corrosion	
Study Goals	The student is able to describe the electrochemical nature of corrosion processes, in his professional life to understand the risks hazards and costs due to corrosion phenomena and act upon it in making decisions on metal applications and to make an argumentative selection of materials classes (steel, stainless steel, aluminium alloys, copper alloys) for given applications. More specifically, the student is able to: 1. compose and interpret electrode potentials and polarisation diagrams for a corroding metal or combination of metals 2. calculate the corrosion rate of metals from electrochemical data set 3. identify and explain the mechanisms of the most important corrosion phenomena 4. select and discuss the technical best protection measures to prevent specific corrosion failures 5. criticize and judge a given materials application	
Education Method	Lectures and guest lectures	
Literature and Study Materials	- Principles and Prevention of Corrosion, Second Edition 1996, van Denny A. Jones, ISBN 0133599930 - lecture slides made available after the lectures	
Assessment	The examination will be as follows: group assignment and an individual closed book written exam. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be: group assignment and a remote open book with several fraud prevention measures.	
Department	3mE Department Materials Science & Engineering	

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Master Materials Science and Engineering

MSE MEA Obligatory selection (Select at least 8 ECTS)

MS43110	Joining Technologies	3
Responsible Instructor	Dr.ir. M.J.M. Hermans	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The course will address a range of joining technologies for different material classes (ceramics, polymers, metals), examining the major features, limitations and applications of selected material process combinations. Attention will be paid to the influence of joining on material properties and the influence of the joint on the structure (where applicable). For the chosen case studies, the selection of joining method and consequences in terms of production and lifetime performance will be assessed and associated failure behaviour will be described.	
Study Goals	At the end of this course, students will be able to: describe the major joining methods available select appropriate joining technique(s) based on the material and application requirements assess the influence of a joining technique on the materials involved describe the main failure mechanisms associated with the different joining techniques considered gain familiarity with working in a team and developing a structured approach to addressing manufacturing issues from a materials science perspective gain experience with written and oral communication skills	
Education Method	Lectures from subject experts and lectures prepared by course participants.	
Assessment	Presented lectures and written examination. Students are required to obtain a minimum mark of 6 for both their lecture and the examination to pass the course.	
Department	3mE Department Materials Science & Engineering	

MS43115	Materials Selection for Engineering Applications	3
Responsible Instructor	Prof.dr.ir. J.M.C. Mol	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Materials in Design - Evolution of Materials Engineering - The Design process - Engineering materials and their properties - Materials selection charts - Materials selection: case studies; Materials processing and design - Materials Processing Selection Charts - Case studies: process selection - Cost considerations - Environmental Impact - Data sources.	
Study Goals	The student is able to develop a systematic procedure for selecting materials and processes with best matches to the requirements of a design More specifically, the student is able to: Integrate materials selection with other aspects of design Develop materials indices and/ or value functions to model design requirements Identify the main materials engineering aspects of materials and process selection. Apply all of the above in case studies representing simplified and real designs.	
Education Method	Lecture	
Computer Use	Cambridge Engineering Selector (CES) Software Package	
Literature and Study Materials	Michael F Ashby Materials Selection in Mechanical Design 5th edition, Butterworth-Heinemann eBook ISBN: 9780081006108 Paperback ISBN: 9780081005996	
Assessment	Lecture notes made available through Brightspace after class Final assignment: Written (remote) report on case study of own choice. This examination format will be independent of unforeseen circumstances or measures resulting from COVID-19.	
Tags	Analysis Design Integrated Optimalisation Process Project Project planning / management Research Methods Sustainability Technology	
Department	3mE Department Materials Science & Engineering	

MS43210	Advanced Characterisation	4
Responsible Instructor	Dr. M.W.E.M. Alfeld	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Required for	MS43800 Determination of Microstructures	
Expected prior knowledge	MS43005 Structure and Properties MS43010 Characterisation of Materials	
Course Contents	This course focusses on microstructure characterization of materials through interaction between x-ray photons, electrons, neutrons and ion beams and solids. This includes the emission of element characteristic x-ray radiation as well as Auger electrons and photoelectrons and phenomena such as: absorption, elastic and inelastic scattering, Sputtering with ions and depth profiling. How these effects are exploited for the characterization of materials will be discussed on the example of several common techniques, such as: electron microscopy, x-ray microanalysis, Auger electron spectroscopy, photoelectron spectroscopy, ion scattering spectroscopy, mass spectrometry and Rutherford backscattering spectroscopy. Further, state of the art synchrotron methods will be discussed. With these, the concepts of depth, spectral and lateral resolution and detection limits will be introduced. Further, common data reduction and evaluation approaches to the high-dimensional hyperspectral data sets acquired by aforementioned techniques will be discussed, as will be the use of tomography for the study of microstructures in 3D volumes. A special focus will be on Orientation Imaging Microscopy (OIM) based on Electron Back Scatter Diffraction (EBSD) and X-Ray Fluorescence (XRF) imaging. The principles of EBSD will be addressed and methods to determine grain size and shape, texture and misorientation between neighbouring grains. For XRF different sources, optics and detectors will be discussed in detail as well as applications of XRF in various field.	
Study Goals	The student is able to apply the physical principles of diffraction and X-ray, electron and ion spectroscopy to explain materials analysis techniques and select suitable techniques for typical problems. More specifically, the student is able to: Describe how the generation of electrons in the elastic and inelastic interaction between electrons and solids, including diffraction, is used to characterize the chemistry and microstructure of materials. Describe how the generation of X-rays in the elastic and inelastic interaction between X-rays and solids, including diffraction, is used to characterize the chemistry and microstructure of materials. Describe how the generation of ions in the elastic and inelastic interaction between ions and solids, is used to characterize the chemistry and microstructure of materials Describe the elastic and inelastic interaction between neutrons and solids is used to characterize the chemistry and microstructure of materials Understand how electron microscopes, diffractometers, and X-ray, electron and mass spectrometers are configured and operated. Describe common data reduction and reconstruction techniques to hyperspectral imaging data sets, as acquired by the techniques discussed. Apply all the above to practical cases in which the chemical composition and microstructure of materials are determined.	
Education Method	Lectures and Assignment	
Literature and Study Materials	Lecture notes	
Books	Microstructural Characterization of Materials, David Brandon, Wayne D. Kaplan, 2008, John Wiley & Sons Ltd., ISBN 0470027851 Fundamentals of Nanoscale Film Analysis, Terry L. Alford, L.C. Feldman, James W. Mayer, Springer, 2007, ISBN 0387292608 Electron Backscatter Diffraction in Materials Science, Adam J. Schwartz, Mukul Kumar, Brent L. Adams, David P. Field, Springer, 2009, ISBN 9780387881355 An Introduction to Synchrotron Radiation, P. Willmott, Wiley, Chichester (UK), 2011, Print ISBN:9780470745793, Online ISBN:9781119970958, DOI:10.1002/9781119970958	
Assessment	The examination will be number of assignments including graded peer-review.	
Elective	Yes	
Tags	Chemistry Image processing Mathematics Physics Technology	
Department	3mE Department Materials Science & Engineering	

MS43801	Determination of Microstructure	2
Responsible Instructor	M.J. Santofimia Navarro	
Instructor	Ing. C. Kwakernaak	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	English	
Expected prior knowledge	MS43210 Advanced Characterisation; it is mandatory to attend these lectures else it will be impossible to do the assignment	
Course Contents	A modern technique to determine the microstructure of crystalline materials is Orientation Imaging Microscopy which is based on Electron Back Scatter Diffraction in a Scanning Electron Microscope. The theory and principles of this technique is presented in the course MS43210 Advanced Characterisation. In this course, however, the theory is put in practise by executing an assignment. Specific skills are trained by applying Orientation Imaging Microscopy in a case study. This course covers fundamental aspects of EBSD measurements, evaluation of collected data and analysis of results. Methods of indexing the diffraction patterns and the determination of crystal orientation. Definitions of coordinate systems and their relation with Euler angles and space. Stereographic projection and representation of crystal orientations in pole figures. Construction of phase and Euler maps. Representation of grain orientations in Rodrigues-Frank space. Determination of misorientation between neighbouring grains. Special orientation relations: coincidence site lattice. Methods to determine grain size and shape, and texture. The assignment involves a laboratory project of an EBSD measurement of a material. A set of diffraction patterns has to be processed. Then, the data set obtained is analysed to determine the microstructure in great detail.	
Study Goals	The student is able to use the physical principles of Electron Back Scatter Diffraction (EBSD) to explain quantitative determination of materials microstructures. More specifically, the student is able to: - Explain the principles of EBSD and the formation of Kikuchi patterns. - Show how electron microscopes equipped with EBSD are built and operated. - Describe how diffraction patterns are sampled, indexed and processed to crystal orientation images. - Formulate how texture of materials is represented using pole figures. - Explain the use of Euler angles and Euler space to characterize crystal orientations. - Compute misorientations and orientation relations between neighboring grains. - Illustrate the representation of EBSD data with the Rodrigues vector in Rodrigues-Frank space. - Describe how orientation maps are constructed and how grain size and shapes are quantified - Apply all the above to analyse the microstructure of a material from an EBSD measurement.	
Education Method	Assignment	
Computer Use	Operation of an electron microscope equipped with EBSD. Processing and analysis of EBSD measurement data.	
Books	Electron Backscatter Diffraction in Materials Science, Adam J. Schwartz, Mukul Kumar, Brent L. Adams, David P. Field, Springer, 2009, ISBN 9780387881355	
Assessment	Oral, including project report and presentation	
Remarks	Old course code: MS3432 and MS43800	
Department	3mE Department Materials Science & Engineering	
Contact	Ing. C. Kwakernaak Phone: +31 15 27 82223 Email: C.Kwakernaak@tudelft.nl Building: 34, room J-1-420	

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Master Materials Science and Engineering

MSE Specialisation Metals Science and Technology

Introduction 1

Metals Science and Technology (MST)

The specialization in Metals Science and Technology will put you in contact with the latest technologies and approaches in the design, characterization, engineering, production and performance of metallic alloys. This specialization will allow you to innovatively design and effectively select metallic alloys able to meet challenging environmental and working conditions. You will earn an in-depth understanding of the relationships between metallic microstructures and their properties, ranging from strength and corrosion to magnetic properties.

This Specialisation Course comprises the following obligatory modules:

-) Science of Failure (MS43100)
-) Metals Science II (MS43200)

Furthermore at least three of the following modules should be taken:

-) Computational Materials Science II (MS43205)
-) Advanced Characterisation (MS43210)
-) Steel Science (MS43215)
-) Corrosion Science (MS43220)

Finally, up to 18 EC of elective modules should be chosen.

Below details of all modules, including the recommended elective modules, of this Specialisation Course can be found.

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Master Materials Science and Engineering

MSE MST Obligatory fixed (7 ECTS)
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MS43100	Science of Failure	3
Responsible Instructor	V. Popovich	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	MS43025 - Mechanical Behaviour of Materials or equivalent knowledge	
Summary	Knowledge, tools and procedures to describe and assess fracture mechanical behaviour of materials.	
Course Contents	Roughly, the outline is as follows (# of lecture hours between brackets): Linear Elastic Fracture Mechanics: stress intensity factor, crack-tip plasticity, energy release rate (4) Elastic Plastic Fracture Mechanics: crack tip opening displacement, J integral (4) Failure: ductile and brittle fracture, ductile-to-brittle transition curve, failure assessment diagram (6) Fatigue: crack growth and crack initiation (4) Environment-assisted cracking: mechanisms and testing (2) Instructions: exercises in class (6) This totals 26 hours.	
Study Goals	The student is able to employ basic fracture mechanical quantities to describe crack growth in solids. Furthermore, the student is able to identify a number of common mechanical phenomena that cause material failure in terms of the mechanisms and the conditions for which such behaviour can be expected. The student understands how to experimentally determine relevant material properties and can make simple failure predictions. More specifically, the student is able to: 1. identify the principles, limitations and applications of the fracture mechanical concepts stress intensity, energy release rate, crack tip opening displacement and J integral; 2. explain experimental procedures to quantify fracture mechanics parameters; 3. identify the major microstructural aspects of brittle and ductile fracture in metals; 4. describe failure assessment in case of ductile and brittle fracture; 5. describe fatigue crack growth with delta K, explain the concept of crack closure, predict fatigue lifetime for constant amplitude loading and illustrate the effect of variable amplitude loading; 6. describe the mechanisms of fatigue crack initiation and relate this to surface roughness and microstructure; 7. illustrate the mechanisms for environment-assisted cracking in metals; 8. explain the principles for quantifying environmentally assisted cracking behaviour and predict lifetime under constant load.	
Education Method	Lectures and instructions	
Literature and Study Materials	Fracture Mechanics, M. Janssen, J. Zuidema and R.J.H. Wanhill, 2nd edition, DUP (2002) Collection of Exercises on Fracture Mechanics, available on Brightspace Slides (including notes) on Brightspace	
Assessment	Written examination during which the book Fracture Mechanics and the hand-outs provided may be consulted. Other forms of information, such as worked-out problems, are not allowed!	
Tags	Industry Mechanics Physics Research Methods Structural Mechanics Sustainability Technology	
Department	3mE Department Materials Science & Engineering	

MS43200	Metals Science II	4
Responsible Instructor	Dr.ir. S.E. Offerman	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	1. Processes of microstructural evolution: diffusional and displacive phase transformations, precipitation, solidification, production routes. The fundamentals of these microstructural processes will be discussed, as well as specific applications for alloys like titanium, aluminium, nickel-based superalloys, shape-memory, super-elastic and invar alloys. 2. The character of various defects in the crystalline structure of metals and mechanisms by which they are introduced into the structure. Defects in the crystal structure can be present according to thermal equilibrium, but density and character can be affected by heat treatments, plastic deformation, high-energy radiation. 3. The basics of various properties of metallic alloys in relation to crystal defects. Properties of metals are primarily based on the crystal structure, but highly dependent on the density and character of crystal defects. Properties that will be discussed include mechanical behaviour, ferromagnetic properties, shape-memory behaviour, temperature resistance.	
Study Goals	From this course the student will acquire knowledge on: 1. the fundamentals of microstructural processes in metals, 2. the introduction, behaviour and effect of defects in crystalline metallic structures, 3. the relation between crystal defects and properties and performance of metals. More specifically, the student will be able to 1. distinguish the effects of diffusion and interfacial processes in diffusional phase transformations and precipitation, 2. identify the main characteristics of solidification of metals, 3. identify the main characteristics of displacive transformations and the differences with diffusional transformations, 4. describe the implications of these mechanisms for various non-ferro alloys, 5. identify the mechanisms by which defects are introduced into metallic crystals, 6. describe processes by which defects in metals can be controlled, 7. describe the influence of crystal defects on magnetic properties of ferromagnetic materials, 8. describe the essentials of shape-memory behaviour, 9. identify the microstructural processes that determine the temperature resistance of metals.	
Education Method	Lectures in interaction with the student	
Assessment	Written exam, open book	
Department	3mE Department Materials Science & Engineering	

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Master Materials Science and Engineering

MSE MST Obligatory selection (Select at least 7 ECTS)

MS43205	Computational Materials Science II	3
Responsible Instructor	Dr. M.H.F. Sluiter	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	computational materials science I (MS43020) and also basics of Newtonian mechanics (3 laws), quantum mechanics (wave function, Schroedinger eqn, uncertainty principle); basics of thermodynamics; calculus; Python, some basic knowledge of algorithms, able to use a computer to run a program and/or script	
Course Contents	Parts 3 and 4 of the course text: Computational Materials Science: Fundamentals to Applications by R. LeSar (ISBN978-0-521-84587-8) Molecular and macromolecular systems Random-walk models of polymers Atomistic simulations of macromolecules Coarse-grained methods Lattice models for polymers and biomolecules Simulations of molecular and macromolecular materials Kinetic Monte Carlo The kinetic Monte Carlo method Time in the kinetic Monte Carlo method Kinetic Monte Carlo calculations Monte Carlo methods at the mesoscale Modeling Grain Growth The Monte Carlo Potts model The N-fold way Applications of the Potts model in materials science and engineering Cellular automata Cellular automata in two dimensions Lattice-gas methods Cellular automata in materials research Relation to Monte Carlo Phase-field methods Conserved and non-conserved order parameters Governing equations A one-dimensional phase-field calculation Free energy of an interface Local free-energy functions Applications in materials science and engineering Mesoscale dynamics Damped dynamics Langevin dynamics Simulation entities at the mesoscale Dynamic models of grain growth Discrete dislocation dynamics simulations Materials selection and design Integrated computational materials engineering Concurrent materials design Materials informatics	
Study Goals	After passing the course, the student will be able to... 1) determine what methods can be joined to model materials phenomena that bridge time and length scales 2) understand the basic principles behind several classes of larger scale models, such as kinetic Monte Carlo for time dependent processes Phase Field models for phase transformations Other mesoscale models such as for grain growth 3) understand, apply, interpret and analyze results from above models 4) apply above methods within "Integrated Computational Materials Engineering" 5) familiarized with basics of Materials Informatics	
Education Method	Lectures & computer exercises (TU Delft computers, Jupyter notebooks with Python)	
Books	Computational Materials Science: Fundamentals to Applications by R. LeSar (ISBN978-0-521-84587-8)	
Prerequisites	computational materials science I (MS43020)	
Assessment	Project. The precise format will be announced at the start of the class.	
Permitted Materials during Tests	TU Delft computer with access to Jupyter notebook server	
Elective	Yes	
Tags	Algebra Algorithms Analysis Artificial intelligence Modelling Numeric Methods Programming Programming concepts Programming Software Research Methods Software Video	
Department	3mE Department Materials Science & Engineering	

MS43210	Advanced Characterisation	4
Responsible Instructor	Dr. M.W.E.M. Alfeld	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Required for	MS43800 Determination of Microstructures	
Expected prior knowledge	MS43005 Structure and Properties MS43010 Characterisation of Materials	
Course Contents	This course focusses on microstructure characterization of materials through interaction between x-ray photons, electrons, neutrons and ion beams and solids. This includes the emission of element characteristic x-ray radiation as well as Auger electrons and photoelectrons and phenomena such as: absorption, elastic and inelastic scattering, Sputtering with ions and depth profiling. How these effects are exploited for the characterization of materials will be discussed on the example of several common techniques, such as: electron microscopy, x-ray microanalysis, Auger electron spectroscopy, photoelectron spectroscopy, ion scattering spectroscopy, mass spectrometry and Rutherford backscattering spectroscopy. Further, state of the art synchrotron methods will be discussed. With these, the concepts of depth, spectral and lateral resolution and detection limits will be introduced. Further, common data reduction and evaluation approaches to the high-dimensional hyperspectral data sets acquired by aforementioned techniques will be discussed, as will be the use of tomography for the study of microstructures in 3D volumes. A special focus will be on Orientation Imaging Microscopy (OIM) based on Electron Back Scatter Diffraction (EBSD) and X-Ray Fluorescence (XRF) imaging. The principles of EBSD will be addressed and methods to determine grain size and shape, texture and misorientation between neighbouring grains. For XRF different sources, optics and detectors will be discussed in detail as well as applications of XRF in various field.	
Study Goals	The student is able to apply the physical principles of diffraction and X-ray, electron and ion spectroscopy to explain materials analysis techniques and select suitable techniques for typical problems. More specifically, the student is able to: Describe how the generation of electrons in the elastic and inelastic interaction between electrons and solids, including diffraction, is used to characterize the chemistry and microstructure of materials. Describe how the generation of X-rays in the elastic and inelastic interaction between X-rays and solids, including diffraction, is used to characterize the chemistry and microstructure of materials. Describe how the generation of ions in the elastic and inelastic interaction between ions and solids, is used to characterize the chemistry and microstructure of materials Describe the elastic and inelastic interaction between neutrons and solids is used to characterize the chemistry and microstructure of materials Understand how electron microscopes, diffractometers, and X-ray, electron and mass spectrometers are configured and operated. Describe common data reduction and reconstruction techniques to hyperspectral imaging data sets, as acquired by the techniques discussed. Apply all the above to practical cases in which the chemical composition and microstructure of materials are determined.	
Education Method	Lectures and Assignment	
Literature and Study Materials	Lecture notes	
Books	Microstructural Characterization of Materials, David Brandon, Wayne D. Kaplan, 2008, John Wiley & Sons Ltd., ISBN 0470027851 Fundamentals of Nanoscale Film Analysis, Terry L. Alford, L.C. Feldman, James W. Mayer, Springer, 2007, ISBN 0387292608 Electron Backscatter Diffraction in Materials Science, Adam J. Schwartz, Mukul Kumar, Brent L. Adams, David P. Field, Springer, 2009, ISBN 9780387881355 An Introduction to Synchrotron Radiation, P. Willmott, Wiley, Chichester (UK), 2011, Print ISBN:9780470745793, Online ISBN:9781119970958, DOI:10.1002/9781119970958	
Assessment	The examination will be number of assignments including graded peer-review.	
Elective	Yes	
Tags	Chemistry Image processing Mathematics Physics Technology	
Department	3mE Department Materials Science & Engineering	

MS43215	Steel Science	3
Responsible Instructor	M.J. Santofimia Navarro	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic knowledge regarding solid-solid phase transformations in metals, such as thermodynamics, diffusion, interfaces, nucleation and growth.	
Course Contents	This course is based on assignments, covering mainly the following topics: - Fe-C equilibrium and plain carbon steels - Fe-C-X equilibrium and alloyed steels - Bainite and acicular ferrite - Formation and properties of martensite - Martensite tempering - Thermomechanical processing - Embrittlement and fracture - Simulations at the micro- and atomic-scale for steel research	
Study Goals	To identify the relations between processing, microstructure and properties of steels.	
Education Method	The lecturers are divided in two parts: - A talk, most of the times by an expert guest lecturer. - Discussion on the assignment expected for that day. The self-study component is given by the timely submission of assignments.	
Literature and Study Materials	Different chapters of the book H.K.D.H. Bhadeshia and R.W.K. Honeycombe Steels: Microstructure and Properties , Butterworth-Heinemann, 4th Edition (2017) will be considered. This book is available online at the library. Relevant scientific articles will be provided in Brightspace.	
Assessment	Assignments on provided scientific articles (80% grade) and presentation (20%). It is important to provide the assignments timely, otherwise the grade of the assignment has a penalty, as indicated in Brightspace. Participation in the class discussions is strongly advised and will be included in the final assessment of the course (+/- 1 point of the final grade). There is no exam.	
Department	3mE Department Materials Science & Engineering	

MS43220	Corrosion Science	3
Responsible Instructor	Y. Gonzalez Garcia	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Corrosion is a multidisciplinary phenomenon in which chemistry, physics, and materials are essential. This course covers the fundamental aspects of corrosion science. Students will gain basic scientific knowledge to understand the corrosion mechanisms. The course will discuss the correlation between microstructural features of the metal and the initiation of corrosion, the formation of passive layers, and the role of the environment in corrosion. The most important corrosion mechanisms will be covered. The main barrier protection methods against corrosion and their failure mechanism will be discussed. Following is a list of the main topics addressed in the course: Introduction to the electrochemistry in corrosion: thermodynamic and kinetic Passivity of the metals Localised corrosion Protection methods against corrosion: pre-treatments and coatings Advanced techniques in corrosion research This course will pay great attention to the scientific aspects of corrosion. Students will have the opportunity to know current research topics and their challenges.	
Study Goals	This course aims to discover more fundamental aspects of the corrosion mechanism of metals and protection methods. In this course, the student will be able to: - understand the thermodynamic and kinetic aspects of the corrosion process - gain a basic understanding of electrochemical methods commonly used in corrosion research - analyse and interpret experimental results - understand the effect of the material properties and the environment on the kinetics of the corrosion process - explain the differences in the passivity of iron, aluminium, and stainless steel - identify the microstructural features of the material that influence the initiation of corrosion of relevant metal alloys - understand and describe the mechanism of localised corrosion failures such as pitting failure - read, review and discuss scientific publications in the corrosion field	
Education Method	The lectures will take place on campus. They will consist of theoretical explanations, discussions, and exercises. For some topics, the students must read and prepare material in advance before the session. The assignment will consist of a critical literature mini-review of a research topic selected by the students (with the instructor's guidance). In the review, students will discuss certain numbers of scientific papers in detail and summarise the essential findings and their relevance to the field. The expected outcome of the assignment is a report and a short presentation about their findings. Depending on the number of students in the course, the assignment will be carried out individually or in small groups.	
Literature and Study Materials	The primary study materials are lecture slides and selected scientific papers. Additionally, some selected chapters will be indicated as support reading material. The chapters are from the following books: - Introduction to Corrosion Science by E. McCafferty Springer 2010 - "Corrosion Engineering" by B. Popov - Science Direct 2015 - "Principles and Prevention of Corrosion" by Denny Jones (2nd edition - 1995)	
Assessment	The assessment consists of the Exam (50% of the final grade) and the written Assignment (40% of the final grade) and the Presentation (10 % of the final grade). To pass the course, the student has to obtain a minimum grade of 6 for each part (exam, assignment and presentation).	
Permitted Materials during Tests	The examination will be on campus. The method of assessment will be a close book written exam. In the case of on-campus restrictions, the exam will be an online written test. In this case, the exam will be open book. You can make use of the lecture notes. The students will be informed at the beginning of the course.	
Department	3mE Department Materials Science & Engineering	
Contact	For general questions or comments about the course, you can contact the docent via email: Dr. Yaiza Gonzalez-Garcia (y.gonzalezgarcia@tudelft.nl).	

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Master Materials Science and Engineering

MSE Specialisation Materials for Sustainable Development

Introduction 1

Materials for Sustainable Development (MSD)

This Specialisation Course is about materials science in the context of sustainable development. A growing world population and increasing welfare urge us to redesign our materials world/system towards a more environmental friendly system. We need to close material loops, keep materials longer in the value chain through re-use and remanufacturing, process materials in a resource efficient way, and design out critical/toxic materials. Another important aspect of this specialisation is the energy transition for the production of electricity from hydrocarbon-fuels towards renewable sources and energy efficiency. Energy generation from renewable sources and energy storage is critically dependent on the availability of materials with appropriate properties.

This Specialisation Course comprises the following obligatory modules:

-) Materials and Sustainable Development (MS43300)
-) Materials for Clean Energy Technologies (MS43305)

Furthermore, at least two of the following modules should be taken:

-) Computational Materials Science II (MS43205)
-) Materials at High Temperature (MS43310)
-) Advanced Characterisation (MS43210)
-) Recycling Engineering Materials (MS43315)

Finally, up to 20 EC of elective modules should be chosen.

Below details of all modules, including the recommended elective modules, of this Specialisation Course can be found.

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Master Materials Science and Engineering

MSE MSD Obligatory fixed (8 ECTS)
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MS43305	Materials for Hydrogen and Solar Energy Technology	4
Responsible Instructor	Dr. A.J. Bottger	
Instructor	Dr. Y. Yang	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	MSE Specialisation: Materials for Sustainable Development	
Expected prior knowledge	Basic materials science knowledge is required. Basics of semi-conductor physics and photo-voltaics is considered as prior knowledge.	
Course Contents	New technologies require new materials. This course starts with a brief overview on the energy consumption, resource constraints and sustainable alternatives. Within that framework the focus of the course is on materials and devices for solar energy and the production and use of hydrogen as an energy carrier. Materials design, manufacturing and materials behaviour under process conditions is addressed.	
Study Goals	After this course the student is able to: 1. explain and evaluate processes and principles used to produce and store hydrogen 2. explain and evaluate processes and principles used to produce solar energy materials 3. analyse and evaluate the effect of the production methods on the energy conversion properties of solar cell material 4. identify, analyse and evaluate the effects of hydrogen gas on materials 5. identify, analyse and evaluate the effects on materials when solar cells are in use. 6. describe the general design of devices used for conversion of solar radiation and heat into electricity 7. critically read, evaluate and discuss recent scientific publications on the course topics.	
Education Method	Class room teaching and self-study (blend learning) or online alternative + self-study	
Literature and Study Materials	Course material: handouts, videos and/or pre-recorded PowerPoint presentations, recent scientific papers and book chapters.	
Assessment	The examination will be as follows: Written or digital open book exam and an oral presentation + discussion; presence during the oral presentation sessions is obligatory. The oral part of the exam (presentation on recent scientific papers) will take place during the last weeks of the course during regular lecture hours. In case of unforeseen circumstances or measures resulting from COVID-19 or other, the prescribed assessment will be: Written open book exam through online assignment and, oral presentation (on recent scientific papers) + discussion in online sessions (Zoom, Teams or other).	
Permitted Materials during Tests	During the written open book exam the use of course material as provided in Brightspace is permitted. The documents will be made available during the exam.	
Department	3mE Department Materials Science & Engineering	
Contact	A.J.Bottger@tudelft	

MS43315	Recycling Engineering Materials	4
Responsible Instructor	Dr. Y. Yang	
Responsible Instructor	S.T. Abrahams	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	General knowledge of materials; general physics; Inorganic chemistry; general thermodynamics	
Course Contents	Recycling of materials from secondary resources is important to assure sustainable materials supply and minimal environmental impact. It is also an indispensable component of circular economy, as it helps close the loop of the whole materials cycle. The main objective of this course is to give students with engineering and science backgrounds a proper understanding of recycling as part of complete materials production and supply chain, a general overview, and insights of available technologies and current practice. The course content is divided into the following topics: Module I: Fundamentals and relevance Introduction: materials recycling as an integrated part of a circular economy Fundamentals of materials recycling Economics of materials recycling Liberation and physical separation technologies Module II: Materials Recycling Technologies Recycling of polymers and plastics: fundamentals and technologies Recycling of polymers and plastics: industrial applications and challenges Recycling of hybrid & composite materials Recycling of metals Module III: Recycling of EoL Engineering applications Recycling of rechargeable batteries Recycling of EoL solar PV panels Recycling of wind turbines & WEEE Recycling of EOL vehicles (ELVs): Industrial practice EOL aircraft management and materials recycling Future strategies: Materials substitution & Design for recyclability	
Study Goals	By the end of this course, you should be able to explain the key steps in the recycling of major engineering materials and design a recycling route for complex engineering applications. In particular, you should be able to: (1) Explain the whole value chain of the materials cycle and the role of recycling to close the loop of the cycle. (2) Describe how major engineering materials are recycled: the best available technologies and industrial practice. (3) Design a recycling route for complex products, and analyze the benefits, challenges, and limitations of the chosen recycling route. (4) Evaluate the economic, social, and environmental impacts of recycling.	
Education Method	The course will be given with the following methods: (1) Lectures on campus (2) Guest lectures from industry experts on campus or at the plant site (3) Industrial excursion to a recycling company (4) A case study (group task, to be assigned during the course, incl. mini MOOC online course) All educational resources will be available on the online platform (BrightSpace). They consist of: (1) Compilation of various book chapters and journal/web articles in digital pdf form (under course materials). (2) Lecture presentations (to be delivered after each lecture) (3) Videos and short movies (to be shown during the lectures), with source information given in the lecture presentations. (4) Optional reading materials (not to be included for examinations)	
Assessment	(1) written exam (combination of fundamental questions and a short essay); (2) case study report and presentation (group work) All assignments are mandatory. Assessment criteria for the assignments are detailed in the course. A case study (in small groups) will be performed by all course participants. It will account for 50% of the total evaluation with a report and a presentation in a mini-symposium. The other 50% will be counted with the written exam (in closed-book normally, or open-book if COVID-19 pandemic restrictions continue).	
Department	3mE Department Materials Science & Engineering	

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Master Materials Science and Engineering

MSE MSD Obligatory selection (Select at least 6 ECTS)

MS43205	Computational Materials Science II	3
Responsible Instructor	Dr. M.H.F. Sluiter	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	computational materials science I (MS43020) and also basics of Newtonian mechanics (3 laws), quantum mechanics (wave function, Schroedinger eqn, uncertainty principle); basics of thermodynamics; calculus; Python, some basic knowledge of algorithms, able to use a computer to run a program and/or script	
Course Contents	Parts 3 and 4 of the course text: Computational Materials Science: Fundamentals to Applications by R. LeSar (ISBN978-0-521-84587-8) Molecular and macromolecular systems Random-walk models of polymers Atomistic simulations of macromolecules Coarse-grained methods Lattice models for polymers and biomolecules Simulations of molecular and macromolecular materials Kinetic Monte Carlo The kinetic Monte Carlo method Time in the kinetic Monte Carlo method Kinetic Monte Carlo calculations Monte Carlo methods at the mesoscale Modeling Grain Growth The Monte Carlo Potts model The N-fold way Applications of the Potts model in materials science and engineering Cellular automata Cellular automata in two dimensions Lattice-gas methods Cellular automata in materials research Relation to Monte Carlo Phase-field methods Conserved and non-conserved order parameters Governing equations A one-dimensional phase-field calculation Free energy of an interface Local free-energy functions Applications in materials science and engineering Mesoscale dynamics Damped dynamics Langevin dynamics Simulation entities at the mesoscale Dynamic models of grain growth Discrete dislocation dynamics simulations Materials selection and design Integrated computational materials engineering Concurrent materials design Materials informatics	
Study Goals	After passing the course, the student will be able to... 1) determine what methods can be joined to model materials phenomena that bridge time and length scales 2) understand the basic principles behind several classes of larger scale models, such as kinetic Monte Carlo for time dependent processes Phase Field models for phase transformations Other mesoscale models such as for grain growth 3) understand, apply, interpret and analyze results from above models 4) apply above methods within "Integrated Computational Materials Engineering" 5) familiarized with basics of Materials Informatics	
Education Method	Lectures & computer exercises (TU Delft computers, Jupyter notebooks with Python)	
Books	Computational Materials Science: Fundamentals to Applications by R. LeSar (ISBN978-0-521-84587-8)	
Prerequisites	computational materials science I (MS43020)	
Assessment	Project. The precise format will be announced at the start of the class.	
Permitted Materials during Tests	TU Delft computer with access to Jupyter notebook server	
Elective	Yes	
Tags	Algebra Algorithmics Analysis Artificial intelligence Modelling Numeric Methods Programming Programming concepts Programming Software Research Methods Software Video	
Department	3mE Department Materials Science & Engineering	

MS43310	Materials at High Temperature	4
Responsible Instructor	V. Popovich	
Instructor	P. Taheri	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Students should already have knowledge on: a) Materials Science b) Mechanical Behaviour c) Basics of Fracture Mechanics d) Fatigue MS43005 - Structure and Properties of Materials covers aspect a). MS43100 - Science of Failure covers aspects b), c) and d).	
Summary	This course relates to materials exposed to high temperatures. It can be divided into two parts: Part A - Mechanical behaviour Part B - High temperature materials, processing and oxidation In the treatment hereof a number of aspects will be addressed: High temperature applications: - combustion engines and gas turbines - boiler materials (heater and steam pipes) - waste and bio mass incinerators - solid oxygen fuel cells Materials for high temperature applications: - Fe and Ni based alloys (Laves phases) - superalloys - intermetallics (TiAl, NiAl etc) - ceramics - refractory materials and coatings - properties-temperature relations Design & processing: - coating deposition - sintering - alloying	
Course Contents	A total of 14 two hour lectures will be taught (number of two hour lectures between brackets): Part A: High Temperature Mechanical Behaviour Creep in crystalline solids (metals and ceramics) (3) - creep testing - creep mechanisms - deformation mechanism maps - extrapolation procedures - creep resistant materials - creep fracture High temperature fatigue (metal alloys) (1) - testing - creep-fatigue interactions - effect of environment Thermomechanical fatigue (metal alloys) (2) - definitions - role of mechanical constraint - elastic / plastic shakedown - thermal ratcheting - testing - TMF in super alloys: effects of phasing, crystal orientation, time, environment - lifetime prediction models Part B: High Temperature oxidation and processing Introduction of high temperature oxidation (1) - cases, applications - methods of investigation - accelerated test - lifetime prediction High temperature processing of thin film materials (1½) - atomic layer deposition (ALD) - chemical vapor deposition (CVD) - physical vapor deposition (PVD) High-temperature surface engineering using (1½) - monolayer deposition (MLD) - high temperature doping - rapid thermal annealing Part C: High Temperature Materials (4) - Ceramics and composites - Thermal barrier coatings - Self-healing in Coatings at High Temperatures - Additive manufacturing and design of high temperature materials In addition to the lectures, exercises will be made and assignments will be given, for example: - prediction of creep lifetime	

Study Goals	- calculation of creep strain rate - assessing thermomechanical fatigue lifetime - assessment of thermal shock behaviour Students are able to: - identify the different mechanisms by which creep can occur in crystalline solids and explain the principle and usage of deformation mechanism maps - explain the principle of superplastic deformation - perform extrapolations of creep rupture data - identify effects of temperature and environment for high-temperature fatigue - describe the conditions for which thermal/thermomechanical fatigue play a role - describe and calculate conditions for which elastic or plastic shakedown occurs - illustrate the conditions for which thermal ratcheting occurs - explain how isothermal and thermomechanical fatigue are tested and how lifetime is predicted - describe possible mechanisms of thermal/thermomechanical fatigue in high-temperature materials and identify the major affecting parameters - describe the phenomenon of thermal shock and explain the major parameters affecting this - understand gas-solid interactions; thermodynamics and reaction kinetics - understand methods of investigation for materials applied at high temperatures - give materials solutions for high temperature applications
Education Method	Lectures, exercises and assignments
Literature and Study Materials	Slides, including notes (on brightspace) Reader (on Brightspace) on the basis of - The ASM Handbook - Deformation and Fracture Mechanics of Engineering Materials, R.W. Hertzberg - Papers and notes on specific applications and cases.
Assessment	Written closed-book exam
Department	3mE Department Materials Science & Engineering

MS43330	Sustainable Metals Production	4
Responsible Instructor	Dr. Y. Yang	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	General metals science; Inorganic chemistry, fundamental physical chemistry (thermodynamics, kinetics)	
Course Contents	Main subjects: Primary and secondary metal resources, extractive metallurgy, metals recycling, pyrometallurgy, hydrometallurgy, electrowinning and electro-refining of metals, ironmaking and steelmaking, decarbonization and green transition of steel industry, extraction of major non-ferrous metals (biref), rare and critical metals, circularity of metals, sustainable development for metals production and recycling, efficient use of energy and materials resource, minimization of environmental footprint (solid & liquid wastes, and CO₂ & other global-warming gas emissions). Lecture topics: The following topics, which are essential for the sustainable metals production, will be covered: Part I: Process fundamentals - theory natural and secondary resources of metals; metals cycle and circularity of metals thermodynamics and kinetics for extractive metallurgy major chemical reactions relevant for the extraction of metals from ore, scrap and waste principles of pyrometallurgy, hydrometallurgy and electrochemical processes - electrometallurgy (electro-winning, electro-refining, molten salt electrolysis) fundamentals of oxide reduction, sulphide smelting, and pyro-refining technologies, with examples of different metals extraction and refining Part II: Production of metals applications and sustainable developments blast furnace ironmaking, alternative and emerging ironmaking technologies with low CO₂ emissions - direct reduction and smelting reduction, hydrogen-based reduction technologies, electrification and electrolysis basic oxygen furnace (BOF) steelmaking; the electric arc furnace (EAF) steelmaking (scrap-based steel production process) secondary steel refining, continuous casting, steel forming brief introduction of major non-ferrous metals production: (1) alumina and aluminium metal production; aluminium recycling; (2) the process routes to produce high purity copper from sulphide and low grade oxide ores (pyrometallurgical route, and hydrometallurgical route), and copper recycling; (3) hydrometallurgical production of zinc: Roasting Leaching Electrowinning (RLE) route; zinc recycling overview on production and recycling of rare and critical metals materials and energy balances, energy and environment, sustainability in metals production	
Study Goals	The students are able to understand and describe the process routes and technologies for producing metals from primary ores and secondary resource (scrap and waste) for the following group of metals: iron and steel, major non-ferrous metals (brief), as well as rare and critical metals (overview). The students can explain the most commonly used methods for producing these metals Pyrometallurgy, Hydrometallurgy, and Electrometallurgy (electrowinning and electrorefining) and understand why these methods are industrially applied, what technologies are involved, which developments are ongoing, and what impacts these processes have on resource utilization, energy consumption and environmental footprint. The students should gain good perception on the technological and societal challenges, and on potential solutions to the sustainable developments for meeting the future growing demands of metals without compromising the environment.	
Education Method	Lectures and guest lectures from industry experts A case study for metallurgical calculations (thermodynamics and energy consumption in ironmaking, kinetic aspects) and process assessments A plant visit/excursion to a production site for ironmaking and steelmaking (Tata Steel, IJmuiden)	
Literature and Study Materials	Selected book chapters from various monographs (listed below) on primary metals production, and metals recycling (copyright eBook and pdf documents). (1) Peter Hayes, Process Principles in Minerals and Materials Production. 4th edition, 2021. Hayes Publishing Co., Australia. (2) Treatise on Process Metallurgy, Volume 3: Industrial Processes. 1st Edition, Editor-in-Chief: Seshadri Seetharaman. 2014 Elsevier. (3) Handbook of Recycling: State-of-the-art for Practitioners, Analysts, and Scientists. Edited by Ernst Worrell and Markus Reuter. 2014 Elsevier. Contemporary literature on metals extraction, refining, and recycling Handouts of the lectures in the form of PowerPoint presentations	
Assessment	Assignment: 50% in the form of case study in small groups Written exam: 50% (closed book)	
Department	3mE Department Materials Science & Engineering	

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Master Materials Science and Engineering

MSE Free Specialisation 14 EC (to be filled in consultation with master coordinator)

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Master Materials Science and Engineering

MSE Recommended electives

AE4ASM001	Design of lightweight structures I: Composites & Metals	3
Responsible Instructor	Dr.ir. O.K. Bergsma	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The global course setup is such that there lectures contain the following topics: - Thoughts behind lightness - Design allowables - Interaction between materials processes - Materials and their properties - Essentials of manufacturing - Processing to final products and their applications. - The principles of stresses in laminated composites - Recycling and other environmental aspects. - The principles of repair of structures. Guest lectures will be illustrating the course content.	
Study Goals	After succeeding this course the student should be able to: - explain parameters and their relationships, which play a role in the development of lightweight structures and parts. Examples: - be able to judge a structural design on conditions required to call a design a lightweight design. - be able to identify carbon glass aramid and dyneema fibres - be able to identify metals from composites on micro and macro scale. - be able to relate lightweight materials to typical strong and weak points in their performance. - be able to give examples of fibre morphologies - be able to argue the correlation between fibre content, orientation control, fibre length, manufacturing process and application. - be able to recall & argue unwanted stress distributions in composite materials	
Education Method	Lectures. The lectures will most likely be given in complete day sections (2 times 2 hours). Collegerama recordings of lectures are available. Online lecture videos are available and explain the content in an other way. Guest lectures are two hour lectures. ALTHOUGH COVID-19 MIGHT INTERFERE, IT IS EXPECTED THAT THE LECTURE WILL BE GIVEN IN A LECTURE ROOM.	
Assessment	Written exam, with grades converted to whole or half numbers. In case of 100 points the following distribution is most often used <points>[grade]: <0-14>[1]; <15>[1.5]; <16-24>[2]; <25>[2.5]; <26-34>[3]; <35>[3.5]; <36-49>[4]; <50>[4.5]; <51-54>[5]; <55>[5.5]; <56-64>[6]; <65>[6.5]; <66-74>[7]; <75>[7.5]; <76-84>[8]; <85>[8.5]; <86-94>[9]; <95>[9.5]; <96-100>[10];	

AE4ASM004	Manufacturing of Aerospace Structures & Materials	3
Responsible Instructor	Ir. J. Sinke	
Responsible Instructor	Dr. K. Masania	
Instructor	I. Fernandez Villegas	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The red line of the course is aiming at knowledge and understanding of manufacturing processes in relation to material properties and feasible product designs. The course contents include manufacturing processes for metallic and composite parts, the assembly of parts into large (sub)structures, and related topics like Quality control, organisation principles, finances, etc.	
Study Goals	The student should have a good knowledge and understanding of the mainstream manufacturing processes of structural materials (lightweight alloys, composites, hybrids). The student should be able to describe and motivate the processing procedures like the processing steps, required tooling and equipment, of manufacturing processes. The student should be able to select adequate manufacturing processes for designs of lightweight structures and components, and should be able to motivate his/her choices. The student should be able to analyse and synthesize interactions between materials, design and manufacturing processes. The student should be able to identify and explain manufacturing related flaws and inaccuracies and advice on how to prevent/limit those.	
Education Method	- Interactive lecture where contents are presented and where the flipped classroom is often used. - Exercises in class with some additional working material	
Assessment	written exam. Closed book.	

AE4ASM005	Fatigue of Structures & Materials	3
Responsible Instructor	Dr.ir. R.C. Alderliesten	
Instructor	Dr.ir. J.A. Pascoe	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	- Introduction to Fatigue (fatigue as a phenomenon; stress concentrations; residual stresses; fatigue properties of metallic and composite materials; fatigue strength of notched specimens, residual strength). - Fatigue damage mechanisms (initiation, crack growth, delamination growth, transverse matrix cracking, fibre failure). - Analysis methods (stress concentration factors, stress intensity factors; energy balance approaches, strain energy release rates). - Fatigue loading (Load Spectra, Fatigue under Constant- & Variable-Amplitude Loading). - Special Fatigue Conditions (surface treatments; fretting corrosion; corrosion fatigue; high-temperature and low-temperature fatigue, moisture ingress). - Fatigue and Damage Tolerance of Aircraft Structures: Regulations, tests, scatter, application of fatigue and damage tolerance methods.	
Study Goals	This course provides the students with engineering knowledge and skills to recognize and to analyse fatigue and damage tolerance problems in aircraft structures and materials. After the course the student must be able 1. Interpret and discuss the fatigue fracture features with respect to the characteristics of each phase in fatigue life 2. Define and determine stress concentration factors for notched structures with or without residual stresses 3. Explain and discuss S-N curves with respect to mean stress, material surface effects, and scatter, and perform fatigue life analyses considering mean stress and notch root plasticity 4. Assess the fatigue life of tension and shear joints, and explain limitations to the similarity principles (K,I,T) 5. Explain Linear Elastic Fracture Mechanics concepts for damage growth, and perform crack growth analyses with these concepts 6. Explain the consequences of variable- and constant amplitude loading on fatigue life and damage growth, and perform fatigue life analyses for arbitrary load spectra 7. Explain the effect of environment on fatigue life and fatigue phenomena 8. Perform residual strength analyses.	
Education Method	Selfstudy using the Online Course Material on Brightspace together with weekly plenary instruction sessions and discussions with fellow (online and campus) students or e-moderator on a forum.	
Literature and Study Materials	J. Schijve, Fatigue of Structures and Materials, 2nd edition, 2009, Springer Science+Business Media, B.V., ISBN-13: 978-1-4020-6807-2 e-ISBN-13: 978-1-4020-6808-9.	
	Readers, formative tests, documentation, and other relevant material provided via Brightspace.	
Assessment	Written exam & assignments Written exam at the end of the lecture period, 80% of total grade. Assignments during the lecture period, altogether 20% of the total grade. Deadlines of individual assignments are communicated via Brightspace.	

AE4ASM103	Functional Coatings	3
Responsible Instructor	Dr. S.J. Garcia Espallargas	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	The students must have general knowledge about materials science.	
Course Contents	(Organic) coatings can be defined as all kind of thin (organic) solid phases mechanically or chemically attached to a much thicker structure in order to add a certain functionality not obtainable from the substrate material itself. Color, mechanical, chemical and corrosion protection are obvious traditional functions of coatings but there are other less classical functions that are quickly growing in importance for several fields including aerospace. This course will introduce the basic principles relevant for traditional organic coatings science (e.g. adhesion, barrier, corrosion, mechanics) and those principles relevant for new developments in functional coatings. Special attention will be put on concepts applicable to aerospace field such as self-healing, active corrosion protection by nano and microparticles, anti-icing, self-cleaning, antifouling and drag reduction. Some of the concepts explained can be extended to other domains closely related to coatings such as adhesives and composites.	
Study Goals	The students will be able to: explain the key mechanisms leading to relevant coating properties and functionalities; use a tool-box to develop new coating systems; relate formulation parameters (chemistry, particles), design and production parameters with final properties; identify suitable techniques to characterize coating properties (e.g. mechanics, viscoelastic properties, barrier, adhesion, crosslinking density); support sound solutions to known problems by analyzing literature.	
Education Method	The course consists of a set of lectures and one assignment followed by a presentation. The assignment is individual. The topic for the assignment will preferentially be unique for each student and related to the course content on coating functionalities. If the number of students registered exceeds a critical number the assignments will be performed in small groups.	
Literature and Study Materials	The lectures (and facilitated slides in pdf) will be supported by selected literature and book chapters that will be facilitated during the lectures. The students will also use specific literature for their assignments.	
Assessment	Written report (40% of final grade) and written exam (60% of final grade). A minimum of a 5 over 10 points will be required in the written exam to make an average with the report.	

AE4ASM109	Design & Analysis of Composite Structures I	5
Responsible Instructor	C. Kassapoglou	
Responsible Instructor	Dr. D. Zarouchas	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Knowledge of Mechanics of Materials Knowledge of Linear Algebra Knowledge of Theory of Elasticity Affinity with infinite series Affinity with Partial Differential Equations Understanding of Experimental Mechanics Programming skills Minimum standards as taught in the BSc. programme of AE. See relevant course descriptions in the AE BSc. study guide	
Course Contents	1. Classical Lamination Theory a. Short Overview of Materials. Composites Design Philosophy b. Theory of Elasticity c. Engineering Constants d. Stress & Strain Transformations / Implications for Testing e. Thin laminates 2. Progressive damage analysis a. Failure criteria b. First ply failure & Last ply failure c. Damage tolerance analysis 3. Reliability analysis & Health monitoring a. Probability of Failure & Uncertainty quantification b. Structural Health Monitoring 4. Basic Stress Solutions and Buckling of Composite Plates a. Typical Airframe Elements b. Airframe Design Process, Materials & Damage c. Plate Governing Equations / Solution of the PDE d. Energy Minimization Methods e. Buckling of Composite Plates	
Study Goals	The students should develop in-depth understanding and insight with regard to the basic mechanics of composite materials and structures, and be able to apply the lectures theories and methods to tackle a variety of basic composite design problems. In addition, the students should become able to expand the provided analysis tools towards more advanced solutions for their graduation thesis work. Furthermore, the students should be able to understand and apply related scientific literature. At the end of this course the student will be able to: - Understand the basic mechanics of composite materials and structures as listed under "course content" - Show insight into the theory of progressive damage analysis of composite materials and structures by applying the theory and methods and tools listed under the course contents to solve a variety of basic composite structure design problems - Understand the philosophy of reliability analysis and calculate the probability of failure of composite structures by combining numerical methods and the mechanics of composite structures - Understand the importance of structural health monitoring (SHM) and design a SHM system for a given composite structure - Demonstrate the ability to expand the theory, methods and tools towards more advanced solutions in real practice such as students may encounter during their thesis - Understand, select on relevance, and apply additional theory, tools and methods on composite materials and structures found in relevant scientific literature to (design) problems based on the material taught in class	
Education Method	Lectures in sessions of 2 X 45 minutes. The distribution in sessions is: 1. Classical Laminate Theory: 3 2. Progressive damage analysis: 3 3. Reliability analysis & Health Monitoring: 4 4. Basic Stress Solutions and Buckling of Comp. Plates: 4	
Literature and Study Materials	Daniel IM, Ishai O. Engineering Mechanics of Composite Materials 2nd Edition, Oxford University Press, 2006. Zarouchas D. Lecture slides on the Classical Lamination Theory, progressive damage analysis, reliability analysis and health monitoring. Available on Brightspace. Kassapoglou C. "Design and Analysis of Composite Structures", 2nd Edition, Wiley, 2013. Kassapoglou C. Lecture slides on Basic Stress Solutions and Buckling of Composite Plates. Available on Brightspace.	
Assessment	Parts A1 and A2 will be examined by means of a take-home assignment. The students should demonstrate correct understanding of the conditions and limitations of the applied methodologies and should prove their ability to solve design problems by the utilization of basic programming techniques and proper results interpretation. They can earn a maximum of 60 points according to the following breakdown: A1. ABD and stress analysis of composite plates 10 points A2. Progressive damage and damage tolerance analysis 20 points A3. Reliability analysis & Health Monitoring. 30 points	

Part B1 will be assessed on the basis a design problem. The design problem will address a realistic composite structure and will require use of all aspects of part (4) as well as use of the Classical Laminated Plate Theory (2) and Failure criteria (3) (see education method).

B1. Design problem 40 points

The maximum score a student can get is thus 100 points. There is no possibility to omit point-related exam parts, for example handing-in a report regarding only parts A1+B1. Every part (Part A1,A2,A3 and part B1) should at least be rewarded with 55% of the attributed points (55%*60 and 55%*40), otherwise the student will fail. In addition, the overall average grade should at least be a 58 on the basis of 100. Grades will be rounded to the nearest half as per regulations.

Students who have handed-in the initial assignments for both parts and have received an overall grade of 5.5 or lower are entitled to a resit assignment in the same academic year. In that case, the student should resit the failed part only. If the student grades higher than 5.5 (pass) and wants to amend the grade, should resit both parts. The date from which the resit assignments are available is announced on Brightspace but will always be during the July-August period.

Each assignment will have a deadline published on Brightspace. Deadlines must be met, no exceptions possible. Assignments are individual and if students work together on a design problem, they must state this in the report naming the collaborators and hand-in individual reports that are 100% original work. (no commonality!). Failure to do so will be seen as plagiarism.

AE4ASM504	Structural Integrity and Maintenance	3
Responsible Instructor	Dr.ir. J.A. Pascoe	
Instructor	Ir. J. Sinke	
Instructor	Dr.ir. R.C. Alderliesten	
Instructor	N. Eleftheroglou	
Instructor	Dr. R.M. Groves	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic knowledge on aircraft structures and materials, and on deterioration of aircraft during operation due to fatigue, corrosion, impact, etc.	
Course Contents	The course is split in a generic part and a part with special topics. For a number of topics, guest lecturers are invited. Generic part - Introduction: Developments in aviation, aircraft technology and safety, effects on Maintenance. Design & safety concepts (safe life, fail safe, damage tolerance). - Maintenance Fundamentals: Definition of maintenance, Maintenance concepts and policies, Failures of airframe structures (Failure definition, Failure types and classes, Degradation types, Reliability, Probability of failures, Relation failure classes and actual failure types, such as fatigue, corrosion, etc.). - Airworthiness Requirements: Main aspects of DT requirements, Fatigue & Damage Tolerance Evaluation (Selection of Critical Locations, Classification of Structure). - Maintenance of Airframe Structures (maintaining structural integrity): ATA-100 system hierarchy, MSG-3 methods (Structural Maintenance Program), Relation between design concepts (e.g. Damage Tolerance), actual design, maintenance and airworthiness requirements. How to translate design results in to maintenance, Structural Repair Manual, Health/condition monitoring (Load/Usage Monitoring & Fleet Life Management). Special Topics (guest lecturers) - Non Destructive Inspections of Airframe Structures: Developments in NDI methods, NDI methods in practical situations. - Load/Usage Monitoring & Fleet Life Management. - Repair of Airframe Structures: Developments in repair methods. - Structural Integrity and Maintenance of Gas Turbines.	
Study Goals	The study goals for this course are: - provide the student knowledge about airworthiness, roles of aerospace stakeholders, maintenance processes during operation, new developments in AC maintenance - the student should understand basic principles in damage tolerance and damage monitoring/inspection, aircraft maintenance - the student should be able to analyse maintenance problems from different angles (technical, regulations, planning). - the student should be able to provide well explained answers/solutions to questions and problems to structural integrity and maintenance issues	
Education Method	Lectures.	
Literature and Study Materials	Lecture notes (hand-outs).	
Assessment	Written exam. Closed book.	

AE4ASM508	Design of Self-healing materials	3
Responsible Instructor	Dr. S.J. Garcia Espallargas	
Instructor	Dr.ir. H.E.J.G. Schlagen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	The students must have general knowledge about materials science.	
Course Contents	Current engineering materials design is based in the well-known concept of damage prevention, which is the improvement of properties necessary to avoid damage. Nevertheless, another concept, employed by natural materials and systems for millions of years, can be applied to man-made engineering materials design. This concept is based on damage management and known as self-repair or self-healing. Under this concept the damage is unavoidable and the systems have to develop mechanisms to autonomously heal the damaged zone to keep the functionality of the system (e.g. skin wound healing). Up to now, material scientists have been able to implement the self-healing concept to polymers, metals, and ceramics in different fields as polymeric and inorganic coatings, high temperature coatings, civil engineering materials as concrete and asphalt, fiber composites, bulk polymers for multipurpose applications, structural aluminium alloys, and others. This course will deal with the self-healing concept, the self-healing materials and systems developed so far, the methods and concepts behind the development of this new class of materials, the techniques employed to evaluate the degree of healing, and the future goals within the self-healing materials field.	
Study Goals	The students will be able to: identify self-healing materials; describe methods to develop self-healing systems; understand the underlying physical and chemical processes behind healing; differentiate advantages and disadvantages of different self-healing approaches; learn how to measure, evaluate and analyze self-healing materials properties.	
Education Method	Lectures and research report. For the research report the students will work in small groups to answer a set of questions related to self-healing materials and their applications.	
Literature and Study Materials	The lectures will be based on the latest developments in the field supported by selected literature. The study material will be the lecture notes and articles facilitated by the instructors.	
Assessment	Oral Exam (40% of final grade) + report (60% of final grade). The report has an individual grade (70%) and a group grade (30%). A minimum grade of 5 over 10 in the exam is mandatory to pass the course.	
Special Information	Assessment may differ depending on the number of registered students.	

AE4ASM510	Design & Analysis of Composite Structures II	3
Responsible Instructor	C. Kassapoglou	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	Skin-stiffened structure: Analysis and Design. Failure modes with emphasis on skin-stiffener separation Sandwich structure: Analysis of different failure modes (buckling including transverse shear, wrinkling, crimping, dimpling). Issues with sandwich rampdown Design guidelines: Robust, manufacturable, and damage tolerant design Effects of damage: Holes, cracks, delaminations, impact. Analysis methods and design guidelines.	
Study Goals	Establish an understanding of how applied loads relate to local stresses and strains in composite structures. Use stresses and strains to predict failure. Combine failure methods into design approaches accounting for different components (skin and stiffeners), design approach (sandwich versus stiffened) and type of damage (hole, crack, delamination, impact)	
Education Method	Lectures	
Books	C. Kassapoglou "Modeling the effect of damage in composite structures", Wiley, 2015	
Prerequisites	AE4ASM109	
Assessment	Homework problems and a final project	

AE4ASM516	Material Selection in Mechanical Design	3
Responsible Instructor	Ir. J. Sinke	
Responsible Instructor	C.D. Rans	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The course aims at training the students in the field of material and process selection on the basis of objective and quantitative criteria. The concept of material selection on the basis of so-called Ashby diagrams will be presented in a number of lectures.	
Study Goals	Providing the student with the tools to make material and process selection on the basis of objective and quantitative criteria.	
Education Method	Lectures	
Literature and Study Materials	The book Materials Selection in Mechanical Design by M.F. Ashby. Butterworth Heinemann. 5th edition 2017	
Assessment	Written exam	
Set-up	The course will consist of a number of lectures in which the quantitative selection process involving an increasing number of constraints and objectives will be introduced. During the lectures students are invited to solve a number of simpler cases.	

BM41035	Biomaterials	4
Responsible Instructor	Dr.ir. I. Apachitei	
Instructor	Dr.ir. E.L. Fratila-Apachitei	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	In improving patient care, especially in relation with tissue loss or dysfunction, the use of proper biomaterials plays a key role. The shift from tissue removal to tissue replacement and at present, tissue regeneration is driven by the (i) evolution of biomaterials from bioinert to bioactive and bioresorbable and (ii) the increasingly complex biomedical problems of an aging and more active population. As an example, the existing total joint replacements have a 75 - 85 % survivability at 15 years. However, since they were first introduced in the 1960s, average life expectancy increased by at least ten years impacting on the quality of the bone and implants survivability that deteriorate with age. As a consequence, improved implant survivability by 10 - 20 years and alternative treatment methods that delay or eliminate the use of prostheses are badly needed. The Biomaterials course covers in an interdisciplinary approach (biomaterial engineering and biology) the background information on type, composition, processing and properties of biomaterials along with their evolution. Metallic, polymeric and ceramic materials are included and their properties are described and compared with those of biological materials. The main methodologies for biomaterials characterization are covered as well as the procedures for biomaterials standardization and regulation. Further, the interactions between biomaterials and specific biological environments are explained and used for defining the specific requirements and selection principles for biomaterials. Here, biomaterials and medical devices for orthopedics, cardiovascular system and tissue engineering are covered. Finally, the research directions and approaches for development of novel biomaterials are outlined.	
Study Goals	At the end of the course the students will be able to: 1. Identify biomaterials used in: orthopedics, cardiovascular implants, and scaffolds for tissue engineering 2. Define the properties of biomaterials and their property map (range of property values) 3. Describe structure-property relationships for various biomaterials 4. Describe the methodologies for biomaterials characterization and regulation 5. Compare and contrast biomaterials and biological materials 6. Explain the main interactions between a biomaterial and a biological environment 7. Describe the principles for biomaterials selection and design 8. Describe approaches and processes used for functionalization of biomaterials 9. Identify, analyse and describe a biomaterial problem associated with a medical device 10. Explain the clinical performance of biomaterials and medical devices	
Education Method	Lectures, case studies and workshops	
Literature and Study Materials	Lecture slides and recommended literature (reader)	
Assessment	Written exam	
Department	3mE Department Biomechanical Engineering	

CH3783	Materials Chemistry for The Nuclear Fuel Cycle	3
Responsible Instructor	A.L. Smith	
Instructor	Dr.ir. S.E. Offerman	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	General chemistry, basics of chemical thermodynamics	
Course Contents	This is a 3 ECTS course composed of 28 total lecture hours, 56 self-study and a final examination. This course is designed for Chemical Engineering, Applied Physics, Materials Science & Engineering, and Sustainable Energy Technology students that are interested in developing a working knowledge of nuclear materials chemistry and the nuclear fuel cycle. The course will cover various aspects of solution and solid state chemistry as well as materials science that play a key role at each step of the nuclear fuel cycle, from the metallurgy of uranium to the disposal of spent reactor fuel or high level waste. While the physics and engineering of controlled fission are central to the generation of electricity by nuclear reactors, chemistry in general and especially materials chemistry dominate all other aspects of the nuclear fuel cycle. This course will give students a comprehensive knowledge of the traditional fuel cycle (the uranium once-through cycle) in Light Water Reactors (LWRs). It will also cover some of the proposed nuclear fuel cycles that may well carry nuclear power through the coming decades. As an outcome of the course, the students will be able to compare and contrast existing and innovative fuel cycles, and discuss the pros and cons of each. In addition, the course will cover some of the unique properties of the actinide 5f-elements. The first three of the actinide series - Th, Pa and U - occur naturally, while the elements following uranium in the Periodic Table (transuranium elements) are man-made. The actinides are essential to nuclear power generation, but also find applications in many other areas of industry, medicine and research. The course will address 5f-electronic configurations, oxidation states, redox potentials, inorganic and structural chemistry, as well thermodynamic properties of actinide compounds. The actinides behaviour in the environment and in the geosphere will be discussed, together with analytical tools for their identification. The biological and environmental hazards associated with these elements pose certain risks which must be controlled and minimized. Due to their significant role and because the final destination of transuranic elements originating from the nuclear fuel cycle is still an open issue, the actinides chemistry and physics continue to be one of the major areas of nuclear research. The materials science part of the course will cover the main features of materials microstructures and behaviour in conditions that are relevant for nuclear reactors. The various types of defects at the atomic or micrometre scale will be explained as well as their consequences in terms of energy. This forms the basis of understanding the changes that materials undergo when subjected to elevated temperature and high-energy radiation. Changes in the microstructure and the defect configurations lead to changes in properties and behaviour and can thus jeopardise the material performance, forming a threat to safety. The course is aimed at master level students with general background in chemistry, and/or materials science.	
Study Goals	1. Students can demonstrate how chemistry and materials science influence almost all aspects of the fuel cycle. 2. Students can describe the key steps involved in the traditional nuclear fuel cycle and explain the involved chemical processes. 3. Students can apply the concepts addressed in the course to discuss alternative fuel cycles, not covered in the class, and explain the fundamental differences with the presented traditional fuel cycle. 4. Students can justify, at each step of the fuel cycle, the behaviour of the different actinide elements in view of their chemical and physical properties (electronic structure, redox potentials, solubility, complexation, speciation, thermodynamics, etc) 5. Students can predict and explain the behaviour of fission products in the irradiated fuel, during the reprocessing and in geological environment. 6. Students are able to predict the environmental behaviour of various actinide species. 7. Students can discuss major analytical techniques for actinide studies. 8. Students can perform numerical evaluations to predict the behaviour of the actinide elements and fission products at each step of the fuel cycle. 9. Students are able to discuss the major features of materials microstructures and behaviour when subjected to the conditions in a nuclear reactor. 10. Students write an essay and make a presentation on a supplemental topic related to the course material, but not covered in class, in which they apply the key concepts presented during the lectures and perform critical analysis of the information.	
Education Method	Oral lectures / Supported self-study	
Literature and Study Materials	Main: Course reader Additional reading: N. Tsoulfanidis. The Nuclear Fuel Cycle, American Nuclear Society, 2013; The Nuclear Fuel Cycle: from Ore to Wastes, P.D. Wilson (ed.), Oxford University Press, 1996. The Chemistry of the Actinide and Transactinide Elements (L.R. Morss, N.M. Edelstein, J. Fuger (Ed.)), Springer Netherlands, 2011 S.A. Cotton, Lanthanide and Actinide Chemistry, John Wiley & Sons, 2006 W. Loveland, D.J. Morrissey, G.T. Seaborg. Modern Nuclear Chemistry, Wiley-Interscience, 2006; G.R. Choppin, J. Rydberg, J.-O. Liljenzin, C. Ekberg, Radiochemistry and Nuclear Chemistry, Academic Press, 2013. Nuclear Fuel Cycle Science and Engineering, I. Crossland (ed.), Woodhead Publishing, 2012 M. Halka, B. Nordstrom, Lanthanides and Actinides, Facts on File Inc., 2011 W. Loveland, D.J. Morrissey, G.T. Seaborg, Modern Nuclear Chemistry, Wiley-Interscience, 2006. G.R. Choppin, J. Rydberg, J.-O. Liljenzin, C. Ekberg, Radiochemistry and Nuclear Chemistry, Academic Press, 2013. William D. Callister Jr., David G. Rethwisch, Materials Science and Engineering: An Introduction, ISBN 978-0470419977, Ed. Wiley	
Assessment	The final grade is composed of the weighted average of the final exam (40%), presentation (20%) and essay (40%). Note: the exact format of assessment may change due to Coronavirus (COVID-19) regulations.	
Elective	Yes	
Tags	Chemistry	
Location	Building 58 (preferred) or Reactor Institute Delft, Radiation Science and Technology department	

ET4376	Photovoltaic Basics	4
Responsible Instructor	Prof.dr.ir. O. Isabella	
Course Coordinator	Dr.ir. R. Santbergen	
Contact Hours / Week x/x/x/x	0/4/0/0 lectures	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course teaches basic knowledge about direct energy conversion of light into electricity that takes place in advanced semiconductor devices called solar cells. Another term that denotes the field of direct solar energy conversion into electricity is Photovoltaics (PV). The course starts with explaining the role of PV technology in energy transition towards sustainable energy. It continues with a short overview of solar cell technologies based on different semiconductor materials, PV markets and applications. The main parameters of radiation spectra, and the solar spectrum in particular, are explained. Suitable solar-cell materials and their optical and electronic properties that are relevant to PV energy conversion, such as absorption of light, transport of charge carries, and their recombination-generation, are discussed. The course focuses on solar-cell operating principles that enable illuminated solar cells to deliver electricity. Dark and illuminated "current-voltage (I-V)" characteristics, solar cell external parameters and their measurement are explained. The course treats the conversion efficiency limits of solar cells. The effect of illumination level, temperature, and parasitic resistances on the performance of solar cells is discussed using the equivalent circuit of a solar cell. The optimal design of solar cells is highlighted based on the analysis and minimization of optical and recombination losses in solar cells. A complete PV system is introduced, the individual components and their characteristics are explained. Practical design and sizing rules of PV systems are exercised.	
Study Goals	Students learn about renewable energy sources, their potential and role in the energy transition towards sustainable energy. The generation of electricity directly from solar energy using solar cells is the focus of the course. This requires understanding the principles of the photovoltaic energy conversion based on the knowledge of material properties, solar cell structures, and processes of absorption of light and transport of charge carriers in the solar cells. Students learn to evaluate the performance limits of different types of solar cells and can design solar cell structures with minimal optical and collection losses in the energy conversion process. The students know the different components of a PV system and understand the design rules of a PV system for various applications.	
Education Method	Lectures, short multiple-question tests, exercises, instructions and self-study using the book Solar Energy.	
Literature and Study Materials	Book Solar Energy: The physics and engineering of photovolataic conversion technologies and systems (Publisher UIT Cambridge 2016), lecture sheets, short instructive videos placed on brightspace, old exams.	
Assessment	Short multiple-question tests, instructions, written exam. All marks will be expressed on a scale of 0 to 10.	
	In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4377	Photovoltaic Technologies	4
Responsible Instructor	Dr. R.A.C.M.M. van Swaaij	
Instructor	Prof.dr. I.M.F. Gordon	
Instructor	Dr. L. Mazzarella	
Contact Hours / Week x/x/x/x	0/0/2/0 lectures; 0/0/2/0 instruction	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Lecture course Photovoltaic Basics (ET4 376)	
Course Contents	This course will focus on several solar-cell technologies: silicon solar cells based on (multi)-crystalline wafers, thin-film solar cells based on thin silicon films, cadmium telluride (CdTe) and copper-indium-(gallium)-(di)selenide (CIS to CIGS), III-V solar cells and concentrators, and organic and perovskite solar cells. For these various technologies the device concepts, the specific properties of the materials required, and the manufacturing and processing methods will be discussed. Before these technologies will be discussed, an in-depth treatment of semiconductor device physics for solar cells and light-management concepts will be given that are applicable to all technologies. Issues included are recombination at interfaces, passivation, heterojunctions, various principles of light-scattering, anti-reflection coatings, and textured interfaces.	
Study Goals	This course aims at an understanding of the different solar-cell technologies. Students are expected to be able to explain the operation of these devices based on the design, processing and material properties. Therefore an understanding of the physics of these technologies is required. The learning objectives (LO) are as follows: LO1: Understand semiconductor processes and light management concepts relevant for solar-cell operation. LO2: Apply the semiconductor processes and light management concepts to existing state-of-the-art and emerging photovoltaic technologies with the objective to understand the solar-cell design. LO3: Analyze the origin of various loss mechanisms in solar cells and the impact of different design solutions to achieve optimal performance.	
Education Method	Flipped-classroom video lectures and on-campus instructions.	
Literature and Study Materials	Arno Smets, Klaus Jäger, Olindo Isabella, René van Swaaij, and Miro Zeman: Solar Energy - The Physics and Engineering of Photovoltaic Conversion Technologies and Systems; UIT Cambridge, England. ISBN: 978 1 906860 32 5 (paperback); This book is also available as a free ebook: ISBN: 978 1 906860 73 8 (ePub) ISBN: 978 1 906860 75 2 (pdf)	
Assessment	The course will be assessed by a combination of a written final examination and three graded non-mandatory assignments. The final mark for this course, F, will be based on the final examination, E, and the average results for the non-mandatory assignments, A. All marks will be expressed on a scale of 0 to 10. For calculating A, the average of the three assignment results is used. Non-mandatory assignments that are not handed in or not handed in on time will have a score of 0. The results for the non-mandatory assignments will be valid only within the academic year in which these assignments have been made. Results for non-mandatory assignments obtained in previous academic years will not be taken into account. The final mark, F, for this course will be computed as follows: $F = E + [A \times (10 - E)/100]$. The computation will be carried out using one decimal and subsequently rounded to halves. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

ID5356	Sustainable Design Strategies for Product Development	3
Course Coordinator	Ir. I.C. de Pauw	
Contact Hours / Week x/x/x/x		
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	In this course you get the chance to practice sustainable product design. You will explore three state-of-the-art strategies for sustainable design and apply these to design a highly sustainable product. We offer this elective because designers can use different strategies to develop sustainable products, but creating a truly beneficial product is a challenge. The course includes Cradle to Cradle, Biomimicry and Ecodesign. The objective is to learn about the strategies, experience applying them and evaluate their effectiveness. If you would like more information about this course, please contact Ingrid de Pauw: i.c.depauw@tudelft.nl.	
Study Goals	After this course you will be able to: - Distinguish the sustainable design strategies Biomimicry, Cradle to Cradle, and Ecodesign, and explain the differences between their core concepts, such as: efficient vs. effective, cradle-to-grave vs. cradle-to-cradle, reductionistic vs. holistic. - Apply the strategies and the associated tools to the design of a sustainable product. - Compare the strategies and critically evaluate their value for industrial design engineers: the core concepts, limitations, the solution space they offer, the specific qualities they bring to the design assignment, and their applicability.	
Education Method	This course is teamwork; in three intensive workshops you explore the sustainable design strategies in detail, and apply them to a design assignment. For each strategy you produce a poster/infographic of your design process and results. In these weeks you'll practice what it means to design a truly green product with each strategy. Every two weeks you report back your findings and receive feedback from the coaches. Please note that your presence during the weekly workshops/feedback sessions is compulsory! The final deliverables are three posters/infographics, a report and a five minute pitch. The final report includes a critical reflection where you compare the strategies and the applicability of these strategies in your future work.	
Assessment	Your report and poster-presentation will be graded, resulting in a combined mark (60% report, 40% posters+presentation). Grading is based on the following criteria: - A high level of critical engagement with the strategies, which should be evident from your assignments, analysis, concept development and reflection on the use of strategies; both in writing and in discussions with your mentors. - Creativity: Show how you applied the strategies and show several realistic, creative solutions for the design problem. - Accurateness: have clear descriptions of choices made and of the process followed, clear calculations (if applicable), unambiguous statements, well-argued interpretations and conclusions. - Group process (individual contributions, teamwork, active presence during course, etc).	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Remarks	Please note that, due to the set-up of this course, your presence during the weekly workshops/feedback sessions is required! So, if you cannot be present during the Friday afternoon sessions, you can not participate in the course.	
Elective	Yes	
Tags	Design Group work Sustainability	

MS43040	Machine Learning for Materials Design	4
Responsible Instructor	Dr. S. Kumar	
Contact Hours / Week x/x/x/x	0.0.4.0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	The graduate-level introductory course on "Machine Learning for Materials Design" will explore the principles, methodologies, and applications of machine learning (ML) in the domain of materials science and engineering. By bridging the gap between traditional materials design approaches and the cutting-edge field of machine learning, we will learn the tools and knowledge necessary for data-driven discovery and development of materials. The following ML aspects are first covered through lectures and live coding: - Introduction to PyTorch - Automatic differentiation - Gradient-based optimization - Best practices in machine learning - Feed-forward neural networks - Convolutional neural networks - Recurrent neural networks - Autoencoding neural networks - Inductive biases Next, a wide range of materials science and engineering applications of the above ML topics will be explored via hands-on and latest research-oriented group projects. Some representative examples include (may vary during the course): - Predicting mechanical behavior of composites and polymers - Identifying molecules with unique properties - Designing metamaterials with tailored mechanical behavior - Characterizing imperfections and defects in additive manufacturing through data and physics	
Study Goals	After successfully completing this course, you will be able to: 1. Understand the fundamentals of machine learning and its application in material science. 2. Develop basic machine learning algorithms based on neural networks 3. Identify appropriate machine learning algorithms based on the data and application in material science context 4. Develop data-driven strategies based on machine learning for new problems	
Education Method	Lectures, exercises, and project. Lectures will involve live coding. Students are asked to bring their own laptops.	
Literature and Study Materials	- Lecture slides and Jupyter workbooks will be provided on Brightspace - No specific textbook - Relevant reading material for each section will be provided	
Prerequisites	IMPORTANT: Good knowledge of linear algebra and familiarity with Python are required. Following are the representative topics that are recommended pre-requisites. Mathematics topics: - Vectors and matrices - Tensors - Coordinate transformations - Differentiation (including of matrices) - Eigenvalues and eigenvectors of matrices - Basic knowledge of probability Python topics: - Creating a Function - Creating a Class and Objects - Plotting and visualization of data - Using Numpy for matrix manipulations - Familiarity with jupyter notebooks Materials topics: - Material selection - Mechanical behaviour of materials	
Assessment	Summative assessment: The assessment will be done based on a project, which will have two components: a final presentation and a report. Both final presentation and report will carry 50% of the final grades each. Formative assessment: During the course, there will be several office hours to provide feedback on the project progress. There will also be a mid-term presentation to discuss the project progress. Additional details will be provided at the beginning of the course.	
Tags	Algorithmics Artificial intelligence Computer Engineering Databases Design Mathematics Mechanics Modelling Numeric Methods Programming Programming concepts Programming Software Project	
Department	3mE Department Materials Science & Engineering	
Contact	Contact by email: Sid.Kumar@tudelft.nl	

MS43100	Science of Failure	3
Responsible Instructor	V. Popovich	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	MS43025 - Mechanical Behaviour of Materials or equivalent knowledge	
Summary	Knowledge, tools and procedures to describe and assess fracture mechanical behaviour of materials.	
Course Contents	Roughly, the outline is as follows (# of lecture hours between brackets): Linear Elastic Fracture Mechanics: stress intensity factor, crack-tip plasticity, energy release rate (4) Elastic Plastic Fracture Mechanics: crack tip opening displacement, J integral (4) Failure: ductile and brittle fracture, ductile-to-brittle transition curve, failure assessment diagram (6) Fatigue: crack growth and crack initiation (4) Environment-assisted cracking: mechanisms and testing (2) Instructions: exercises in class (6) This totals 26 hours.	
Study Goals	The student is able to employ basic fracture mechanical quantities to describe crack growth in solids. Furthermore, the student is able to identify a number of common mechanical phenomena that cause material failure in terms of the mechanisms and the conditions for which such behaviour can be expected. The student understands how to experimentally determine relevant material properties and can make simple failure predictions. More specifically, the student is able to: 1. identify the principles, limitations and applications of the fracture mechanical concepts stress intensity, energy release rate, crack tip opening displacement and J integral; 2. explain experimental procedures to quantify fracture mechanics parameters; 3. identify the major microstructural aspects of brittle and ductile fracture in metals; 4. describe failure assessment in case of ductile and brittle fracture; 5. describe fatigue crack growth with delta K, explain the concept of crack closure, predict fatigue lifetime for constant amplitude loading and illustrate the effect of variable amplitude loading; 6. describe the mechanisms of fatigue crack initiation and relate this to surface roughness and microstructure; 7. illustrate the mechanisms for environment-assisted cracking in metals; 8. explain the principles for quantifying environmentally assisted cracking behaviour and predict lifetime under constant load.	
Education Method	Lectures and instructions	
Literature and Study Materials	Fracture Mechanics, M. Janssen, J. Zuidema and R.J.H. Wanhill, 2nd edition, DUP (2002) Collection of Exercises on Fracture Mechanics, available on Brightspace Slides (including notes) on Brightspace	
Assessment	Written examination during which the book Fracture Mechanics and the hand-outs provided may be consulted. Other forms of information, such as worked-out problems, are not allowed!	
Tags	Industry Mechanics Physics Research Methods Structural Mechanics Sustainability Technology	
Department	3mE Department Materials Science & Engineering	

MS43110	Joining Technologies	3
Responsible Instructor	Dr.ir. M.J.M. Hermans	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The course will address a range of joining technologies for different material classes (ceramics, polymers, metals), examining the major features, limitations and applications of selected material process combinations. Attention will be paid to the influence of joining on material properties and the influence of the joint on the structure (where applicable). For the chosen case studies, the selection of joining method and consequences in terms of production and lifetime performance will be assessed and associated failure behaviour will be described.	
Study Goals	At the end of this course, students will be able to: describe the major joining methods available select appropriate joining technique(s) based on the material and application requirements assess the influence of a joining technique on the materials involved describe the main failure mechanisms associated with the different joining techniques considered gain familiarity with working in a team and developing a structured approach to addressing manufacturing issues from a materials science perspective gain experience with written and oral communication skills	
Education Method	Lectures from subject experts and lectures prepared by course participants.	
Assessment	Presented lectures and written examination. Students are required to obtain a minimum mark of 6 for both their lecture and the examination to pass the course.	
Department	3mE Department Materials Science & Engineering	

MS43115	Materials Selection for Engineering Applications	3
Responsible Instructor	Prof.dr.ir. J.M.C. Mol	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Materials in Design - Evolution of Materials Engineering - The Design process - Engineering materials and their properties - Materials selection charts - Materials selection: case studies; Materials processing and design - Materials Processing Selection Charts - Case studies: process selection - Cost considerations - Environmental Impact - Data sources.	
Study Goals	The student is able to develop a systematic procedure for selecting materials and processes with best matches to the requirements of a design More specifically, the student is able to: - Integrate materials selection with other aspects of design - Develop materials indices and/ or value functions to model design requirements - Identify the main materials engineering aspects of materials and process selection. - Apply all of the above in case studies representing simplified and real designs.	
Education Method	Lecture	
Computer Use	Cambridge Engineering Selector (CES) Software Package	
Literature and Study Materials	Michael F Ashby Materials Selection in Mechanical Design 5th edition, Butterworth-Heinemann eBook ISBN: 9780081006108 Paperback ISBN: 9780081005996 Lecture notes made available through Brightspace after class	
Assessment	Final assignment: Written (remote) report on case study of own choice. This examination format will be independent of unforeseen circumstances or measures resulting from COVID-19.	
Tags	Analysis Design Integrated Optimalisation Process Project Project planning / management Research Methods Sustainability Technology	
Department	3mE Department Materials Science & Engineering	

MS43120	Corrosion Engineering	3
Responsible Instructor	Prof.dr.ir. J.M.C. Mol	
Co-responsible for assignments	P. Taheri	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Summary	corrosion, corrosion principles, corrosion types, corrosion prevention and protection	
Course Contents	- Relevance of corrosion - Costs to society - Definitions and electrochemical character of corrosion - General corrosion vs local forms of corrosion - Electrochemical Thermodynamics - Electrochemical Kinetics - Passivity - Galvanic Corrosion and intergranular corrosion - Pitting and Crevice Corrosion - Protection against corrosion	
Study Goals	The student is able to describe the electrochemical nature of corrosion processes, in his professional life to understand the risks hazards and costs due to corrosion phenomena and act upon it in making decisions on metal applications and to make an argumentative selection of materials classes (steel, stainless steel, aluminium alloys, copper alloys) for given applications. More specifically, the student is able to: 1. compose and interpret electrode potentials and polarisation diagrams for a corroding metal or combination of metals 2. calculate the corrosion rate of metals from electrochemical data set 3. identify and explain the mechanisms of the most important corrosion phenomena 4. select and discuss the technical best protection measures to prevent specific corrosion failures 5. criticize and judge a given materials application	
Education Method	Lectures and guest lectures	
Literature and Study Materials	- Principles and Prevention of Corrosion, Second Edition 1996, van Denny A. Jones, ISBN 0133599930 - lecture slides made available after the lectures	
Assessment	The examination will be as follows: group assignment and an individual closed book written exam. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be: group assignment and a remote open book with several fraud prevention measures.	
Department	3mE Department Materials Science & Engineering	

MS43200	Metals Science II	4
Responsible Instructor	Dr.ir. S.E. Offerman	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	1. Processes of microstructural evolution: diffusional and displacive phase transformations, precipitation, solidification, production routes. The fundamentals of these microstructural processes will be discussed, as well as specific applications for alloys like titanium, aluminium, nickel-based superalloys, shape-memory, super-elastic and invar alloys. 2. The character of various defects in the crystalline structure of metals and mechanisms by which they are introduced into the structure. Defects in the crystal structure can be present according to thermal equilibrium, but density and character can be affected by heat treatments, plastic deformation, high-energy radiation. 3. The basics of various properties of metallic alloys in relation to crystal defects. Properties of metals are primarily based on the crystal structure, but highly dependent on the density and character of crystal defects. Properties that will be discussed include mechanical behaviour, ferromagnetic properties, shape-memory behaviour, temperature resistance.	
Study Goals	From this course the student will acquire knowledge on: 1. the fundamentals of microstructural processes in metals, 2. the introduction, behaviour and effect of defects in crystalline metallic structures, 3. the relation between crystal defects and properties and performance of metals. More specifically, the student will be able to 1. distinguish the effects of diffusion and interfacial processes in diffusional phase transformations and precipitation, 2. identify the main characteristics of solidification of metals, 3. identify the main characteristics of displacive transformations and the differences with diffusional transformations, 4. describe the implications of these mechanisms for various non-ferro alloys, 5. identify the mechanisms by which defects are introduced into metallic crystals, 6. describe processes by which defects in metals can be controlled, 7. describe the influence of crystal defects on magnetic properties of ferromagnetic materials, 8. describe the essentials of shape-memory behaviour, 9. identify the microstructural processes that determine the temperature resistance of metals.	
Education Method	Lectures in interaction with the student	
Assessment	Written exam, open book	
Department	3mE Department Materials Science & Engineering	

MS43205	Computational Materials Science II	3
Responsible Instructor	Dr. M.H.F. Sluiter	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	computational materials science I (MS43020) and also basics of Newtonian mechanics (3 laws), quantum mechanics (wave function, Schroedinger eqn, uncertainty principle); basics of thermodynamics; calculus; Python, some basic knowledge of algorithms, able to use a computer to run a program and/or script	
Course Contents	Parts 3 and 4 of the course text: Computational Materials Science: Fundamentals to Applications by R. LeSar (ISBN978-0-521-84587-8) Molecular and macromolecular systems Random-walk models of polymers Atomistic simulations of macromolecules Coarse-grained methods Lattice models for polymers and biomolecules Simulations of molecular and macromolecular materials Kinetic Monte Carlo The kinetic Monte Carlo method Time in the kinetic Monte Carlo method Kinetic Monte Carlo calculations Monte Carlo methods at the mesoscale Modeling Grain Growth The Monte Carlo Potts model The N-fold way Applications of the Potts model in materials science and engineering Cellular automata Cellular automata in two dimensions Lattice-gas methods Cellular automata in materials research Relation to Monte Carlo Phase-field methods Conserved and non-conserved order parameters Governing equations A one-dimensional phase-field calculation Free energy of an interface Local free-energy functions Applications in materials science and engineering Mesoscale dynamics Damped dynamics Langevin dynamics Simulation entities at the mesoscale Dynamic models of grain growth Discrete dislocation dynamics simulations Materials selection and design Integrated computational materials engineering Concurrent materials design Materials informatics	
Study Goals	After passing the course, the student will be able to... 1) determine what methods can be joined to model materials phenomena that bridge time and length scales 2) understand the basic principles behind several classes of larger scale models, such as kinetic Monte Carlo for time dependent processes Phase Field models for phase transformations Other mesoscale models such as for grain growth 3) understand, apply, interpret and analyze results from above models 4) apply above methods within "Integrated Computational Materials Engineering" 5) familiarized with basics of Materials Informatics	
Education Method	Lectures & computer exercises (TU Delft computers, Jupyter notebooks with Python)	
Books	Computational Materials Science: Fundamentals to Applications by R. LeSar (ISBN978-0-521-84587-8)	
Prerequisites	computational materials science I (MS43020)	
Assessment	Project. The precise format will be announced at the start of the class.	
Permitted Materials during Tests	TU Delft computer with access to Jupyter notebook server	
Elective	Yes	
Tags	Algebra Algorithms Analysis Artificial intelligence Modelling Numeric Methods Programming Programming concepts Programming Software Research Methods Software Video	
Department	3mE Department Materials Science & Engineering	

MS43210	Advanced Characterisation	4
Responsible Instructor	Dr. M.W.E.M. Alfeld	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Required for	MS43800 Determination of Microstructures	
Expected prior knowledge	MS43005 Structure and Properties MS43010 Characterisation of Materials	
Course Contents	This course focusses on microstructure characterization of materials through interaction between x-ray photons, electrons, neutrons and ion beams and solids. This includes the emission of element characteristic x-ray radiation as well as Auger electrons and photoelectrons and phenomena such as: absorption, elastic and inelastic scattering, Sputtering with ions and depth profiling. How these effects are exploited for the characterization of materials will be discussed on the example of several common techniques, such as: electron microscopy, x-ray microanalysis, Auger electron spectroscopy, photoelectron spectroscopy, ion scattering spectroscopy, mass spectrometry and Rutherford backscattering spectroscopy. Further, state of the art synchrotron methods will be discussed. With these, the concepts of depth, spectral and lateral resolution and detection limits will be introduced. Further, common data reduction and evaluation approaches to the high-dimensional hyperspectral data sets acquired by aforementioned techniques will be discussed, as will be the use of tomography for the study of microstructures in 3D volumes. A special focus will be on Orientation Imaging Microscopy (OIM) based on Electron Back Scatter Diffraction (EBSD) and X-Ray Fluorescence (XRF) imaging. The principles of EBSD will be addressed and methods to determine grain size and shape, texture and misorientation between neighbouring grains. For XRF different sources, optics and detectors will be discussed in detail as well as applications of XRF in various field.	
Study Goals	The student is able to apply the physical principles of diffraction and X-ray, electron and ion spectroscopy to explain materials analysis techniques and select suitable techniques for typical problems. More specifically, the student is able to: Describe how the generation of electrons in the elastic and inelastic interaction between electrons and solids, including diffraction, is used to characterize the chemistry and microstructure of materials. Describe how the generation of X-rays in the elastic and inelastic interaction between X-rays and solids, including diffraction, is used to characterize the chemistry and microstructure of materials. Describe how the generation of ions in the elastic and inelastic interaction between ions and solids, is used to characterize the chemistry and microstructure of materials Describe the elastic and inelastic interaction between neutrons and solids is used to characterize the chemistry and microstructure of materials Understand how electron microscopes, diffractometers, and X-ray, electron and mass spectrometers are configured and operated. Describe common data reduction and reconstruction techniques to hyperspectral imaging data sets, as acquired by the techniques discussed. Apply all the above to practical cases in which the chemical composition and microstructure of materials are determined.	
Education Method	Lectures and Assignment	
Literature and Study Materials	Lecture notes	
Books	Microstructural Characterization of Materials, David Brandon, Wayne D. Kaplan, 2008, John Wiley & Sons Ltd., ISBN 0470027851 Fundamentals of Nanoscale Film Analysis, Terry L. Alford, L.C. Feldman, James W. Mayer, Springer, 2007, ISBN 0387292608 Electron Backscatter Diffraction in Materials Science, Adam J. Schwartz, Mukul Kumar, Brent L. Adams, David P. Field, Springer, 2009, ISBN 9780387881355 An Introduction to Synchrotron Radiation, P. Willmott, Wiley, Chichester (UK), 2011, Print ISBN:9780470745793, Online ISBN:9781119970958, DOI:10.1002/9781119970958	
Assessment	The examination will be number of assignments including graded peer-review.	
Elective	Yes	
Tags	Chemistry Image processing Mathematics Physics Technology	
Department	3mE Department Materials Science & Engineering	

MS43215	Steel Science	3
Responsible Instructor	M.J. Santofimia Navarro	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic knowledge regarding solid-solid phase transformations in metals, such as thermodynamics, diffusion, interfaces, nucleation and growth.	
Course Contents	This course is based on assignments, covering mainly the following topics: - Fe-C equilibrium and plain carbon steels - Fe-C-X equilibrium and alloyed steels - Bainite and acicular ferrite - Formation and properties of martensite - Martensite tempering - Thermomechanical processing - Embrittlement and fracture - Simulations at the micro- and atomic-scale for steel research	
Study Goals	To identify the relations between processing, microstructure and properties of steels.	
Education Method	The lecturers are divided in two parts: - A talk, most of the times by an expert guest lecturer. - Discussion on the assignment expected for that day. The self-study component is given by the timely submission of assignments.	
Literature and Study Materials	Different chapters of the book H.K.D.H. Bhadeshia and R.W.K. Honeycombe Steels: Microstructure and Properties , Butterworth-Heinemann, 4th Edition (2017) will be considered. This book is available online at the library. Relevant scientific articles will be provided in Brightspace.	
Assessment	Assignments on provided scientific articles (80% grade) and presentation (20%). It is important to provide the assignments timely, otherwise the grade of the assignment has a penalty, as indicated in Brightspace. Participation in the class discussions is strongly advised and will be included in the final assessment of the course (+/- 1 point of the final grade). There is no exam.	
Department	3mE Department Materials Science & Engineering	

MS43220	Corrosion Science	3
Responsible Instructor	Y. Gonzalez Garcia	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Corrosion is a multidisciplinary phenomenon in which chemistry, physics, and materials are essential. This course covers the fundamental aspects of corrosion science. Students will gain basic scientific knowledge to understand the corrosion mechanisms. The course will discuss the correlation between microstructural features of the metal and the initiation of corrosion, the formation of passive layers, and the role of the environment in corrosion. The most important corrosion mechanisms will be covered. The main barrier protection methods against corrosion and their failure mechanism will be discussed. Following is a list of the main topics addressed in the course: Introduction to the electrochemistry in corrosion: thermodynamic and kinetic Passivity of the metals Localised corrosion Protection methods against corrosion: pre-treatments and coatings Advanced techniques in corrosion research This course will pay great attention to the scientific aspects of corrosion. Students will have the opportunity to know current research topics and their challenges.	
Study Goals	This course aims to discover more fundamental aspects of the corrosion mechanism of metals and protection methods. In this course, the student will be able to: - understand the thermodynamic and kinetic aspects of the corrosion process - gain a basic understanding of electrochemical methods commonly used in corrosion research - analyse and interpret experimental results - understand the effect of the material properties and the environment on the kinetics of the corrosion process - explain the differences in the passivity of iron, aluminium, and stainless steel - identify the microstructural features of the material that influence the initiation of corrosion of relevant metal alloys - understand and describe the mechanism of localised corrosion failures such as pitting failure - read, review and discuss scientific publications in the corrosion field	
Education Method	The lectures will take place on campus. They will consist of theoretical explanations, discussions, and exercises. For some topics, the students must read and prepare material in advance before the session. The assignment will consist of a critical literature mini-review of a research topic selected by the students (with the instructor's guidance). In the review, students will discuss certain numbers of scientific papers in detail and summarise the essential findings and their relevance to the field. The expected outcome of the assignment is a report and a short presentation about their findings. Depending on the number of students in the course, the assignment will be carried out individually or in small groups.	
Literature and Study Materials	The primary study materials are lecture slides and selected scientific papers. Additionally, some selected chapters will be indicated as support reading material. The chapters are from the following books: - Introduction to Corrosion Science by E. McCafferty Springer 2010 - "Corrosion Engineering" by B. Popov - Science Direct 2015 - "Principles and Prevention of Corrosion" by Denny Jones (2nd edition - 1995)	
Assessment	The assessment consists of the Exam (50% of the final grade) and the written Assignment (40% of the final grade) and the Presentation (10 % of the final grade). To pass the course, the student has to obtain a minimum grade of 6 for each part (exam, assignment and presentation).	
Permitted Materials during Tests	The examination will be on campus. The method of assessment will be a close book written exam. In the case of on-campus restrictions, the exam will be an online written test. In this case, the exam will be open book. You can make use of the lecture notes. The students will be informed at the beginning of the course.	
Department	3mE Department Materials Science & Engineering	
Contact	For general questions or comments about the course, you can contact the docent via email: Dr. Yaiza Gonzalez-Garcia (y.gonzalezgarcia@tudelft.nl).	

MS43305	Materials for Hydrogen and Solar Energy Technology	4
Responsible Instructor	Dr. A.J. Bottger	
Instructor	Dr. Y. Yang	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	MSE Specialisation: Materials for Sustainable Development	
Expected prior knowledge	Basic materials science knowledge is required. Basics of semi-conductor physics and photo-voltaics is considered as prior knowledge.	
Course Contents	New technologies require new materials. This course starts with a brief overview on the energy consumption, resource constraints and sustainable alternatives. Within that framework the focus of the course is on materials and devices for solar energy and the production and use of hydrogen as an energy carrier. Materials design, manufacturing and materials behaviour under process conditions is addressed.	
Study Goals	After this course the student is able to: 1. explain and evaluate processes and principles used to produce and store hydrogen 2. explain and evaluate processes and principles used to produce solar energy materials 3. analyse and evaluate the effect of the production methods on the energy conversion properties of solar cell material 4. identify, analyse and evaluate the effects of hydrogen gas on materials 5. identify, analyse and evaluate the effects on materials when solar cells are in use. 6. describe the general design of devices used for conversion of solar radiation and heat into electricity 7. critically read, evaluate and discuss recent scientific publications on the course topics.	
Education Method	Class room teaching and self-study (blend learning) or online alternative + self-study	
Literature and Study Materials	Course material: handouts, videos and/or pre-recorded PowerPoint presentations, recent scientific papers and book chapters.	
Assessment	The examination will be as follows: Written or digital open book exam and an oral presentation + discussion; presence during the oral presentation sessions is obligatory. The oral part of the exam (presentation on recent scientific papers) will take place during the last weeks of the course during regular lecture hours. In case of unforeseen circumstances or measures resulting from COVID-19 or other, the prescribed assessment will be: Written open book exam through online assignment and, oral presentation (on recent scientific papers) + discussion in online sessions (Zoom, Teams or other).	
Permitted Materials during Tests	During the written open book exam the use of course material as provided in Brightspace is permitted. The documents will be made available during the exam.	
Department	3mE Department Materials Science & Engineering	
Contact	A.J.Bottger@tudelft	

MS43310	Materials at High Temperature	4
Responsible Instructor	V. Popovich	
Instructor	P. Taheri	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Students should already have knowledge on: a) Materials Science b) Mechanical Behaviour c) Basics of Fracture Mechanics d) Fatigue MS43005 - Structure and Properties of Materials covers aspect a). MS43100 - Science of Failure covers aspects b), c) and d).	
Summary	This course relates to materials exposed to high temperatures. It can be divided into two parts: Part A - Mechanical behaviour Part B - High temperature materials, processing and oxidation In the treatment hereof a number of aspects will be addressed: High temperature applications: - combustion engines and gas turbines - boiler materials (heater and steam pipes) - waste and bio mass incinerators - solid oxygen fuel cells Materials for high temperature applications: - Fe and Ni based alloys (Laves phases) - superalloys - intermetallics (TiAl, NiAl etc) - ceramics - refractory materials and coatings - properties-temperature relations Design & processing: - coating deposition - sintering - alloying	
Course Contents	A total of 14 two hour lectures will be taught (number of two hour lectures between brackets): Part A: High Temperature Mechanical Behaviour Creep in crystalline solids (metals and ceramics) (3) - creep testing - creep mechanisms - deformation mechanism maps - extrapolation procedures - creep resistant materials - creep fracture High temperature fatigue (metal alloys) (1) - testing - creep-fatigue interactions - effect of environment Thermomechanical fatigue (metal alloys) (2) - definitions - role of mechanical constraint - elastic / plastic shakedown - thermal ratcheting - testing - TMF in super alloys: effects of phasing, crystal orientation, time, environment - lifetime prediction models Part B: High Temperature oxidation and processing Introduction of high temperature oxidation (1) - cases, applications - methods of investigation - accelerated test - lifetime prediction High temperature processing of thin film materials (1½) - atomic layer deposition (ALD) - chemical vapor deposition (CVD) - physical vapor deposition (PVD) High-temperature surface engineering using (1½) - monolayer deposition (MLD) - high temperature doping - rapid thermal annealing Part C: High Temperature Materials (4) - Ceramics and composites - Thermal barrier coatings - Self-healing in Coatings at High Temperatures - Additive manufacturing and design of high temperature materials In addition to the lectures, exercises will be made and assignments will be given, for example: - prediction of creep lifetime	

Study Goals	- calculation of creep strain rate - assessing thermomechanical fatigue lifetime - assessment of thermal shock behaviour Students are able to: - identify the different mechanisms by which creep can occur in crystalline solids and explain the principle and usage of deformation mechanism maps - explain the principle of superplastic deformation - perform extrapolations of creep rupture data - identify effects of temperature and environment for high-temperature fatigue - describe the conditions for which thermal/thermomechanical fatigue play a role - describe and calculate conditions for which elastic or plastic shakedown occurs - illustrate the conditions for which thermal ratcheting occurs - explain how isothermal and thermomechanical fatigue are tested and how lifetime is predicted - describe possible mechanisms of thermal/thermomechanical fatigue in high-temperature materials and identify the major affecting parameters - describe the phenomenon of thermal shock and explain the major parameters affecting this - understand gas-solid interactions; thermodynamics and reaction kinetics - understand methods of investigation for materials applied at high temperatures - give materials solutions for high temperature applications
Education Method	Lectures, exercises and assignments
Literature and Study Materials	Slides, including notes (on brightspace) Reader (on Brightspace) on the basis of - The ASM Handbook - Deformation and Fracture Mechanics of Engineering Materials, R.W. Hertzberg - Papers and notes on specific applications and cases.
Assessment	Written closed-book exam
Department	3mE Department Materials Science & Engineering

MS43315	Recycling Engineering Materials	4
Responsible Instructor	Dr. Y. Yang	
Responsible Instructor	S.T. Abrahami	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	General knowledge of materials; general physics; Inorganic chemistry; general thermodynamics	
Course Contents	Recycling of materials from secondary resources is important to assure sustainable materials supply and minimal environmental impact. It is also an indispensable component of circular economy, as it helps close the loop of the whole materials cycle. The main objective of this course is to give students with engineering and science backgrounds a proper understanding of recycling as part of complete materials production and supply chain, a general overview, and insights of available technologies and current practice. The course content is divided into the following topics: Module I: Fundamentals and relevance Introduction: materials recycling as an integrated part of a circular economy Fundamentals of materials recycling Economics of materials recycling Liberation and physical separation technologies Module II: Materials Recycling Technologies Recycling of polymers and plastics: fundamentals and technologies Recycling of polymers and plastics: industrial applications and challenges Recycling of hybrid & composite materials Recycling of metals Module III: Recycling of EoL Engineering applications Recycling of rechargeable batteries Recycling of EoL solar PV panels Recycling of wind turbines & WEEE Recycling of EOL vehicles (ELVs): Industrial practice EOL aircraft management and materials recycling Future strategies: Materials substitution & Design for recyclability	
Study Goals	By the end of this course, you should be able to explain the key steps in the recycling of major engineering materials and design a recycling route for complex engineering applications. In particular, you should be able to: (1) Explain the whole value chain of the materials cycle and the role of recycling to close the loop of the cycle. (2) Describe how major engineering materials are recycled: the best available technologies and industrial practice. (3) Design a recycling route for complex products, and analyze the benefits, challenges, and limitations of the chosen recycling route. (4) Evaluate the economic, social, and environmental impacts of recycling.	
Education Method	The course will be given with the following methods: (1) Lectures on campus (2) Guest lectures from industry experts on campus or at the plant site (3) Industrial excursion to a recycling company (4) A case study (group task, to be assigned during the course, incl. mini MOOC online course) All educational resources will be available on the online platform (BrightSpace). They consist of: (1) Compilation of various book chapters and journal/web articles in digital pdf form (under course materials). (2) Lecture presentations (to be delivered after each lecture) (3) Videos and short movies (to be shown during the lectures), with source information given in the lecture presentations. (4) Optional reading materials (not to be included for examinations)	
Assessment	(1) written exam (combination of fundamental questions and a short essay); (2) case study report and presentation (group work) All assignments are mandatory. Assessment criteria for the assignments are detailed in the course. A case study (in small groups) will be performed by all course participants. It will account for 50% of the total evaluation with a report and a presentation in a mini-symposium. The other 50% will be counted with the written exam (in closed-book normally, or open-book if COVID-19 pandemic restrictions continue).	
Department	3mE Department Materials Science & Engineering	

MS43325	Application of Materials in High Tech Engineering	3
Responsible Instructor	Dr.ir. S.E. Offerman	
Instructor	Dr. E.M. Kelder	
Instructor	Dr. E. Mendes	
Instructor	V. Popovich	
Instructor	Dr.ir. M.J.M. Hermans	
Instructor	Dr. H.L. Castricum	
Contact Hours / Week x/x/x/x	0.4.0.0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basics of Materials Science and Engineering. Preferably, the BSc-course 'Materials Science and Engineering' at 3mE, TU Delft.	
Course Contents	The course aims at training students in performing material and process selection by using objective and quantitative criteria. A substantial part of the course consists of a compilation of actual cases from and presented by the Dutch High Tech industry (ASML, Cerasec, Euro-Techniek, Hittech, Ramlab, Thermo Fisher, VDL ETG, Tata Steel). TU lecturers subsequently provide further insight into the related materials science background and selection method. Cases encompass: additive manufacturing, steel and metal alloys, titanium, metallic glasses, coatings, ceramics, polymers. Themes: Materials in design - Engineering materials and their properties - High tech applications - Materials selection - Process selection - Case studies - Relation microstructure / technical properties / processing / design - Cost Considerations - Environmental Impact - Data sources.	
Study Goals	Understand the importance of and attain insight into material selection in the design and production of High Tech products, systems and components therein. Acquire the tools to carry out material and process selection based on objective and quantitative criteria. Acquire experience-based knowledge of applications and materials used in High Tech industry. Obtain a firmer knowledge base in the associated material science theory. Acquire adequate basic knowledge for one's future role as an academic engineer in industry: being able to make a good first selection of materials, as well as to understand and apply the advice of materials experts and processing experts. Being able to critically assess both industrial experience and advice from materials overviews and databases.	
Education Method	Case presentations; lectures	
Computer Use	Ansys Granta EduPack Software Package	
Literature and Study Materials	Michael F Ashby Materials Selection in Mechanical Design 5th edition, Butterworth-Heinemann eBook ISBN: 9780081006108 Paperback ISBN: 9780081005996 W.D. Callister & D.G. Rethwisch Materials Science & Engineering ISBN:978-1-118-31922-2 Reader background materials science for the case studies of the high-tech industry	
Assessment	Digital exam	
Department	3mE Department Materials Science & Engineering	

MS43330	Sustainable Metals Production	4
Responsible Instructor	Dr. Y. Yang	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	General metals science; Inorganic chemistry, fundamental physical chemistry (thermodynamics, kinetics)	
Course Contents	Main subjects: Primary and secondary metal resources, extractive metallurgy, metals recycling, pyrometallurgy, hydrometallurgy, electrowinning and electro-refining of metals, ironmaking and steelmaking, decarbonization and green transition of steel industry, extraction of major non-ferrous metals (biref), rare and critical metals, circularity of metals, sustainable development for metals production and recycling, efficient use of energy and materials resource, minimization of environmental footprint (solid & liquid wastes, and CO₂ & other global-warming gas emissions). Lecture topics: The following topics, which are essential for the sustainable metals production, will be covered: Part I: Process fundamentals - theory natural and secondary resources of metals; metals cycle and circularity of metals thermodynamics and kinetics for extractive metallurgy major chemical reactions relevant for the extraction of metals from ore, scrap and waste principles of pyrometallurgy, hydrometallurgy and electrochemical processes - electrometallurgy (electro-winning, electro-refining, molten salt electrolysis) fundamentals of oxide reduction, sulphide smelting, and pyro-refining technologies, with examples of different metals extraction and refining Part II: Production of metals applications and sustainable developments blast furnace ironmaking, alternative and emerging ironmaking technologies with low CO₂ emissions - direct reduction and smelting reduction, hydrogen-based reduction technologies, electrification and electrolysis basic oxygen furnace (BOF) steelmaking; the electric arc furnace (EAF) steelmaking (scrap-based steel production process) secondary steel refining, continuous casting, steel forming brief introduction of major non-ferrous metals production: (1) alumina and aluminium metal production; aluminium recycling; (2) the process routes to produce high purity copper from sulphide and low grade oxide ores (pyrometallurgical route, and hydrometallurgical route), and copper recycling; (3) hydrometallurgical production of zinc: Roasting Leaching Electrowinning (RLE) route; zinc recycling overview on production and recycling of rare and critical metals materials and energy balances, energy and environment, sustainability in metals production	
Study Goals	The students are able to understand and describe the process routes and technologies for producing metals from primary ores and secondary resource (scrap and waste) for the following group of metals: iron and steel, major non-ferrous metals (brief), as well as rare and critical metals (overview). The students can explain the most commonly used methods for producing these metals Pyrometallurgy, Hydrometallurgy, and Electrometallurgy (electrowinning and electrorefining) and understand why these methods are industrially applied, what technologies are involved, which developments are ongoing, and what impacts these processes have on resource utilization, energy consumption and environmental footprint. The students should gain good perception on the technological and societal challenges, and on potential solutions to the sustainable developments for meeting the future growing demands of metals without compromising the environment.	
Education Method	Lectures and guest lectures from industry experts A case study for metallurgical calculations (thermodynamics and energy consumption in ironmaking, kinetic aspects) and process assessments A plant visit/excursion to a production site for ironmaking and steelmaking (Tata Steel, IJmuiden)	
Literature and Study Materials	Selected book chapters from various monographs (listed below) on primary metals production, and metals recycling (copyright eBook and pdf documents). (1) Peter Hayes, Process Principles in Minerals and Materials Production. 4th edition, 2021. Hayes Publishing Co., Australia. (2) Treatise on Process Metallurgy, Volume 3: Industrial Processes. 1st Edition, Editor-in-Chief: Seshadri Seetharaman. 2014 Elsevier. (3) Handbook of Recycling: State-of-the-art for Practitioners, Analysts, and Scientists. Edited by Ernst Worrell and Markus Reuter. 2014 Elsevier. Contemporary literature on metals extraction, refining, and recycling Handouts of the lectures in the form of PowerPoint presentations	
Assessment	Assignment: 50% in the form of case study in small groups Written exam: 50% (closed book)	
Department	3mE Department Materials Science & Engineering	

MS43801	Determination of Microstructure	2
Responsible Instructor	M.J. Santofimia Navarro	
Instructor	Ing. C. Kwakernaak	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	English	
Expected prior knowledge	MS43210 Advanced Characterisation; it is mandatory to attend these lectures else it will be impossible to do the assignment	
Course Contents	A modern technique to determine the microstructure of crystalline materials is Orientation Imaging Microscopy which is based on Electron Back Scatter Diffraction in a Scanning Electron Microscope. The theory and principles of this technique is presented in the course MS43210 Advanced Characterisation. In this course, however, the theory is put in practise by executing an assignment. Specific skills are trained by applying Orientation Imaging Microscopy in a case study. This course covers fundamental aspects of EBSD measurements, evaluation of collected data and analysis of results. Methods of indexing the diffraction patterns and the determination of crystal orientation. Definitions of coordinate systems and their relation with Euler angles and space. Stereographic projection and representation of crystal orientations in pole figures. Construction of phase and Euler maps. Representation of grain orientations in Rodrigues-Frank space. Determination of misorientation between neighbouring grains. Special orientation relations: coincidence site lattice. Methods to determine grain size and shape, and texture. The assignment involves a laboratory project of an EBSD measurement of a material. A set of diffraction patterns has to be processed. Then, the data set obtained is analysed to determine the microstructure in great detail.	
Study Goals	The student is able to use the physical principles of Electron Back Scatter Diffraction (EBSD) to explain quantitative determination of materials microstructures. More specifically, the student is able to: - Explain the principles of EBSD and the formation of Kikuchi patterns. - Show how electron microscopes equipped with EBSD are built and operated. - Describe how diffraction patterns are sampled, indexed and processed to crystal orientation images. - Formulate how texture of materials is represented using pole figures. - Explain the use of Euler angles and Euler space to characterize crystal orientations. - Compute misorientations and orientation relations between neighboring grains. - Illustrate the representation of EBSD data with the Rodrigues vector in Rodrigues-Frank space. - Describe how orientation maps are constructed and how grain size and shapes are quantified - Apply all the above to analyse the microstructure of a material from an EBSD measurement.	
Education Method	Assignment	
Computer Use	Operation of an electron microscope equipped with EBSD. Processing and analysis of EBSD measurement data.	
Books	Electron Backscatter Diffraction in Materials Science, Adam J. Schwartz, Mukul Kumar, Brent L. Adams, David P. Field, Springer, 2009, ISBN 9780387881355	
Assessment	Oral, including project report and presentation	
Remarks	Old course code: MS3432 and MS43800	
Department	3mE Department Materials Science & Engineering	
Contact	Ing. C. Kwakernaak Phone: +31 15 27 82223 Email: C.Kwakernaak@tudelft.nl Building: 34, room J-1-420	

MS43805	Material and Fabrication Factors in Marine Structures	5
Responsible Instructor	Y. Gonzalez Garcia	
Instructor	Dr.ir. M.J.M. Hermans	
Instructor	Dr.ir. R.H. Petrov	
Contact Hours / Week x/x/x/x	0/0/0/5	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course is focused on the application of metals in marines structures. Attention is paid to the structure and properties of steels and light alloys as well as their corrosion resistance. The weldability of high-strength steels is addressed and factors influencing weld joint design and design for welding are considered, together with assessment considerations. The fundamental aspects of corrosion mechanism, corrosion monitoring and control will be introduced. The general outline of the course (number of lectures): - Basics of metal structures, phase diagram, phase transformation of metals (2) - Influence of alloying elements on steel structures (1) - Steel production basics (1) - Rolling technology (2) - Al and Al alloys (1) - Welding process & Solidification (2) - Joining dissimilar materials (2) - Additive manufacturing (1) - Corrosion: Thermodynamic and Kinetics (2) - Corrosion types (1) - Corrosion steels and Light-alloys (1) - Corrosion Protection and Control (1) - Cathodic protection (2)	
Course Contents Continuation	The group assignment will focus on a case selected by the students. Based on the specific requirements of the chosen case, the group must (i) recommend the most appropriate material(s) and production procedure, (ii) assess the risk for mechanical failures and propose possible mitigation technique(s) and (iii) evaluate the corrosion susceptibility and recommend method(s) for protection. Each group will deliver a written report. Example of cases: - A helideck on an offshore platform - The boom of a high lift capacity offshore crane - A new oil pipeline to be laid at 3-km water-depth - An lightweight alloy ship hull - Liquefied natural gas storage tank	
Study Goals	- Describe the main factors governing the structure and properties of metals - Identify joint designs typically employed in fabrication - Determine design factors influencing welded joint design and joint performance - Describe corrosion principles and the correlation with material properties - Identify the main corrosion types frequently found in marine structures - Determine appropriate corrosion protection methods based on their underlying behaviour and mechanisms - Select criteria for corrosion control programs	
Education Method	Lectures and recommended reading	
Literature and Study Materials	The metal fabrication part of the course is addressed through readers. The corrosion part of the course is based on selected book chapters that will be available during the course.	
Assessment	Examination (60%) and Assignment (40%). The exam is in closed-book written format. Students are required to have a minimum score of 6 in both the examination and the assignment to pass this course.	
Department	3mE Department Materials Science & Engineering	

MS43810	Materials in Art and Design	3
Responsible Instructor	Prof.dr. J. Dik	
Instructor	Dr. M.W.E.M. Alfeld	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Summary	Culture, art, archaeology, history, degradation, authenticity	
Course Contents	In recent years materials science is increasingly applied to the study of art and archaeological objects. This field is known as archaeometry and it includes three research directions. First of all, authenticity studies on art objects rely on material analysis of the object. Second, artworks suffer from degradation studies that need to be understood at a materials science level. Third, stylistic developments in art history can often be related to the development of new materials or production techniques. The main challenge in this field is to bridge the interface between the humanities and the sciences. Interpreting the findings of materials science in a non-scientific, historical context is the crucial skill in this course. Besides literature study and lecturing, the course includes excursions to art and archaeological collections.	
Study Goals	The students is able to: - give a general overview of archaeometric approaches in material cultural heritage - identify the archaeometric component of archaeological or art historical research questions - describe relevant analytical techniques in archaeometry - interpret analytical data in the archaeological and art historical context and discuss the findings with specialists from the cultural heritage domain	
Education Method	Lecture	
Assessment	Presentation	
Remarks	Old course code: MS4071 Laboratory work: depending on the case different analytical techniques may be applied (light microscopy, electron microscopy, x-ray powder diffraction).	
Department	3mE Department Materials Science & Engineering	

MS43825	Radiation Damage in Materials	4
Responsible Instructor	Dr.ir. S.E. Offerman	
Contact Hours / Week x/x/x/x	0.0.4.0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic materials science (e.g. WB2330: Materials Science of BSc-ME 3mE): point defects, dislocations, grain boundaries, phases, precipitates, thermodynamics and kinetics, diffusion, relationship microstructure and mechanical properties.	
Course Contents	This course describes the effects of irradiation of mainly metals & alloys and explains the mechanisms behind those effects, especially for neutron radiation. The first part explains how damage is created, quantified analytically, and simulated. The differences and similarities between damage created by various particles is discussed. This forms the basic understanding for the physical, microstructural and mechanical changes metals and alloys undergo due to radiation. The 2nd and 3rd part of the courses discusses radiation-induced physical, microstructural and mechanical effects qualitatively and quantitatively. This course is designed for Material Science & Engineering, Applied Physics, and Chemical Engineering students, who are interested in material changes resulting from particle irradiation (mainly neutrons) of materials (mainly metals and alloys) in nuclear and fusion reactors. Lectures will be given by (guest-)lecturers from NRG, DIFFER and TU Delft. Visits to NRG and DIFFER are part of this course. Some of the lectures will be given at NRG and DIFFER.	
Study Goals	1. Students learn the difference in defect morphology between ion and neutron irradiation 2. Students can analytically quantify the amount of damage introduced by various particles 3. Students can describe the key effects, which lead to physical, microstructural and mechanical changes of irradiated materials 4. Students can perform numerical calculations to physical, microstructural and mechanical changes in metals	
Education Method	Oral lectures / Supported self-study / Homework / Site visit	
Literature and Study Materials	Fundamentals of Radiation Materials Science - Metals and Alloys", second edition, by Gary S. Was, Springer 2007, ISBN 978-1-4939-3436-2, ISBN 978-1-4939-3438-6 (eBook), DOI 10.1007/978-1-4939-3438-6	
Assessment	Open book, written exam	
Department	3mE Department Materials Science & Engineering	

SET3055	Economics and Regulations of Sustainable Energy Systems	4
Responsible Instructor	Dr. S.T.H. Storm	
Instructor	Prof.dr.ir. L.J. de Vries	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	None	
Course Contents	This course is a general introduction to energy economics. In this course, you learn the essentials of how actors in the energy system make economic decisions, when markets work well and when they may fail, and what options the government has to remedy market failures. Specifically, the following topics are addressed: Introduction to energy (electricity) markets: perfect competition; oligopoly Rational (and irrational) market behaviour; ABM ? Investment decision making; LCOE Opportunity costing / discounting; NPV; IRR; discount rate Markets, (perfect) competition and market failure (esp. externalities & lock-in) Private costs, external cost and social cost; empirical evidence (related to energy) LCOE; discounting; private discount rate; IRR and NPV Social cost-benefit analysis; social discount rate	
Study Goals	After this course students are able to: evaluate and analyse the performance of energy (electricity) markets; identify cases of market failures and design remedial action to rectify these market failures; estimate the LCOE and calculate the NPV and IRR of an electricity investment project; contrast private and social costs of (renewable) electricity generation; assess technological lock-in in energy generation; and appraise the role of public policy in energy markets and in energy technology innovation.	
Education Method	This course consists of lectures. Parallel to the course a simulation game will be played by the students.	
Literature and Study Materials	To be announced.	
Assessment	The final grade of the course consists of the following components: - Individual computer exam (80%) - Report on the simulation game (20%) Students should score at least 6 for both parts. All marks will be expressed on a scale of 1 to 10. Final grade calculation: $(0.8 * \text{computer exam}) + (0.2 * \text{report})$ In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Computer exam: resit computer exam - Report on the simulation: resubmission of the report Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	G.O. Ndubuisi	

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Master Materials Science and Engineering

Master Year 2

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Master Materials Science and Engineering

MSE MSc Projects and Assignments

MS43702	Materials Elective Project I	2
Responsible Instructor	Y. Gonzalez Garcia	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	Different, to be announced	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	This elective course (2 ECs) allows students to study a selected aspect (as a project) of Materials Science and Engineering in greater depth than is covered in the general MSE masters curriculum. The project can be literature-based, laboratory-based, or numerically based and should lead to an enhanced understanding of the selected topic. The project will not be permitted as part of a students course list unless approved a priori. The MSE coordinator (MSEcoordinator-3mE@tudelft.nl) will assess the relevance and applicability of the proposed project to the MSE programme and the suitability and academic level of the proposed supervisor. The MSE master coordinator must approve proposals for projects (and supervisors) BEFORE being undertaken by the student. The student will gain experience performing an investigative task and applying a critical attitude to the methodology and outcomes of the work. The project will be assessed based on a scientific written report and will be marked by the project supervisor (MSE staff member) using the corresponding grading matrix. The MSE education coordinator will check and, if necessary, moderate the mark before submission in Osiris. The elective is only open to students studying within the Materials Science and Engineering Department. This course cannot be on the same topic as Materials Elective Project I (MS43702) or the External Research Project (MS53200).	
Study Goals	At the conclusion of this project, students should be able to: - apply knowledge at the Materials Science and Engineering master level to the definition solution and discussion of a relevant problem within the framework of the topic selected; - apply relevant knowledge at different levels of abstraction; - show critical analysis of the methods, results and interpretation employed during the project; - communicate effectively regarding solutions to problems, conclusions and knowledge gained with professionals in the field.	
Education Method	Individual work	
Assessment	Grade of the written scientific report Specific grading matrix (available in the Brightspace of Materials Science and Engineering)	
Department	3mE Department Materials Science & Engineering	

MS43706	Materials Elective Project II	6
Responsible Instructor	Y. Gonzalez Garcia	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	Different, to be announced	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	This elective course (6 ECs) allows students to study a selected aspect (as a project) of Materials Science and Engineering in greater depth than is covered in the general MSE masters curriculum. The project can be literature-based, laboratory-based or numerically based and should lead to an enhanced understanding of the selected topic. The project will not be permitted as part of a students course list unless approved a priori. The MSE coordinator (MSEcoordinator-3mE@tudelft.nl) will assess the relevance and applicability of the proposed project to the MSE programme and the suitability and academic level of the proposed supervisor. The MSE master coordinator must approve proposals for projects (and supervisors) BEFORE being undertaken by the student. The student will gain experience performing an investigative task and applying a critical attitude to the methodology and outcomes of the work. The project will be assessed based on a written report (or scientific paper) and will be marked by the project supervisor (MSE staff member) using the corresponding grading matrix. The MSE education coordinator will check and, if necessary, moderate the mark before submission in Osiris. The elective is only open to students studying within the Materials Science and Engineering Department. This course cannot be on the same topic as Materials Elective Project I (MS43702) or the External Research Project (MS53200).	
Study Goals	At the conclusion of this project, students should be able to: - apply knowledge at the Materials Science and Engineering master level to the definition solution and discussion of a relevant problem within the framework of the topic selected; - apply relevant knowledge at different levels of abstraction; - show critical analysis of the methods, results and interpretation employed during the project; - communicate efficiently in the style of a journal or conference article.	
Education Method	Individual work	
Assessment	Grade of the written scientific report or paper in the style of a published article Specific grading matrix (available in the Brightspace of Materials Science and Engineering)	
Department	3mE Department Materials Science & Engineering	

MS53200	External Research Project	15
Responsible Instructor	Y. Gonzalez Garcia	
Instructor	Prof.dr.ir. J.M.C. Mol	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	Different, to be announced	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	The External Research Project involves a research assignment performed individually during which the student works (more or less) full-time in an external research environment, whether academic or industrial. The assignment is at a level close to that expected during the Masters research project. In this way, the student learns to analyse a given issue, devise an approach toward a solution, set up and perform measurements or calculations and discuss and conclude from the results. A report is written that describes and discusses the work performed. By performing the External Research Project, the student learns to: - define, limit, solve and discuss problems within the framework of a predefined project both independently and in consultation with others; - report on the work performed - gain experience in an organisation other than TU Delft. This project must be on a different topic than MS43702 Materials Elective Project I or MS43706 Materials Elective Project II. The course is only open to students studying within the Materials Science and Engineering Department.	
Study Goals	At the conclusion of this study, the student is able to: - utilise a broad and profound knowledge of Materials Science and Engineering in the definition solution and discussion of a relevant problem within the framework of a project addressing engineering sciences (mathematics, physics and chemistry); - apply relevant knowledge at different levels of abstraction, including a reflective understanding of relationships to other fields. - apply knowledge of paradigms, methods and tools, to critically analyse, model, simulate, and/or design in the performance of the research concerning the defined problem. - demonstrate capabilities to solve technological problems systematically, involving critical problem analysis, formulating sub-problems and providing innovative technical solutions in new and unfamiliar situations. - work independently and/or in multidisciplinary teams (as the situation demands), interacting with specialists and taking the initiative where necessary. - communicate effectively regarding solutions to problems, conclusions, and knowledge gained with professionals in the field.	
Education Method	Individual research project	
Practical Guide	The contact between the student and host institution can be initiated by either the student, the host institution, the External Research Project coordinator or another academic staff member. The host institution agrees on an assignment with the student that requires an academic approach on the level reached after the first year of the Masters programme. A brief assignment description is submitted to the coordinator beforehand for approval. The responsibility for the assignment and the supervision is with one of the lecturers of Delft University of Technology, chosen by the student and the coordinator in relation to the subject of the assignment. The host organisation also assigns a supervisor. The student will work most of the time at the host organisation, but, if it is beneficial for the project, part of the time can be spent at Delft University. The External Research Project is finalised by a written report, which is submitted to the supervisor at the host organisation, the supervising lecturer at the university and the coordinator of the External Research Project within four weeks of completion of the practical work. The report should have a typical length of 20 to 40 pages, be written in English and give an account of the approach, the results obtained, a critical discussion of the results and any conclusions drawn. The report should also contain a concise description of the relevance of the project for the host institution. The assessment of the External Research Project is undertaken by the co-ordinator on the basis of indications by the supervisors at the host institute and the supervising lecturer of the university.	
Assessment	Written Report The student will receive a final grade (6 is the minimum passing grade)	
Department	3mE Department Materials Science & Engineering	

TUD4040	Joint Interdisciplinary Project	15
Responsible Instructor	Prof.dr.ir. J. Hellendoorn	
Project Coordinator	Ir. B.J.E. de Bruin	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Summary	JIP consists of three sets of activities: 1. The project which takes place at a company. Students are responsible for the project work (collaboration, integration of different perspectives, the content quality of their work). They plan their activities via a scrum method, are challenged to realise field trips, consult with experts and realise the necessary research work. The team keeps a Scrum log, run a blog for interaction with the outside world and every team member keeps a personal development log. 2. Meeting Professionals lectures and workshops about specialised topics, by academic staff or senior professionals from the companies involved. 3. Plenary meetings and design reviews for all the teams, company coaches and academic staff. At the meetings the students present their intermediate resp. final outcomes and plans, and are provided with feedback.	
Course Contents	The aim of the Joint Interdisciplinary Project is to prepare students to contribute to solving impactful technological challenges. The projects not only demand good engineering working knowledge but also experience with interdisciplinary and systems theory, and both knowledge and mindsets of innovation and entrepreneurial behavior. The project brief is provided by renowned companies like Airbus, Arcadis, etc. Teams of interdisciplinary student teams guided by a company coach and offered academic and industry expertise, are invited to realise an innovative problem solution to a complex problem and contributing to the sustainable development goals.	
Study Goals	1. Cognitive abilities attributable to interdisciplinary learning Demonstrate the ability to engage in perspective-taking; Develop structural knowledge pertaining to the problem; Integrate knowledge and modes of thinking drawn from two or more disciplines; Produce an interdisciplinary understanding of complex problem or intellectual question. 2. Scientific and intellectual development Capable to analyze scientific and societal consequences (economic, social, cultural, environmental) of the innovation; 3. Research and design capabilities Demonstrate engineering skills: technical skills, interpreting results, creativity, usability for company/institute; Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; 4. Collaboration and communication in an interdisciplinary team Demonstrate behavioral competences and skills: taking initiative, responsibility, showing communication skills, independency, collaboration and the ability to respect different disciplines and adapt to different cultures); Show ability to write a technical report: structured/consistent, language proficient, with correct use of literature/references, use of figures/tables/equations, and has a concise format (30 pages); Present work performed in a structured way through an oral presentation to their peers and customer. 5. Self-adjustment and reflection capabilities Plan and control the project efficiently considering resources and methodology; Being able to reflect on personal functioning in an evaluation report: reflect on personal objectives, indicate personal strengths/weaknesses. Indicate future personal improvement, drawing conclusions for future career. Cognitive abilities attributable to interdisciplinary learning: The ability to integrate (scientific and practical technological) knowledge from different disciplines to solve complex problems Scientific and intellectual development The capacity to evaluate the ethical, scientific and societal consequences of the proposed innovation Research and design capabilities The ability to create reasonable and relevant research or design , according to the academic standards of the involved disciplines Collaboration and communication in an interdisciplinary team Demonstrate behavioural competences and skills relevant for teamwork and effective communication with different stakeholders Self-adjustment and reflection capabilities To carry out regular reflections on professional and personal development and being able to improve upon those reflections Understand contemporary and societal issues in their work.	
Education Method	Full-time project work in an interdisciplinary team of about five students. The project work is interspersed with some just in time seminars and workshops about specialized subjects, methods or practical situations that play an essential role in interdisciplinary project work.	
Assessment	The assessment criteria pertain to the process, product and presentation and are assessed at an individual and team level, during 3 review sessions at three levels of accomplishment (each review assesses at a different level): Interdisciplinary work Scientific reasoning and ethical mindset Innovation process Presentation and communication Academic staff, supported by the input from company coaches, grade the students during the Problem Statement, the Midterm and the Final Review. The final group mark for JIP is differentiated per person and is based on: The team presentations - 30% The Problem definition, progress report and final report 30% The individual contribution to the team 20% The final individual reflection report 5% The Blog 5%	
Remarks	Project Outcomes (product, model, system) are presented at: 1. Problem Statement Review 2. Midterm meeting 3. Final Review 4. Symposium presentation	

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Master Materials Science and Engineering

MSE MSc Graduation Project (45 ECTS)

Introduction 1	Literature Research and Master Thesis Project During this final master programme component the individual student undertakes a literature research (10 EC) and performs independent scientific research leading to a thesis (35 EC). The Master Thesis Project should only be undertaken, if no more than 10 EC from other courses are still to be obtained. Furthermore, before starting the project the student should make sure that one of the staff members of the department Materials Science and Engineering takes responsibility for it and subsequently acts as a thesis supervisor. The Master Thesis Project may be related to the chosen Specialisation Course, but can also concern another Materials Science and Engineering subject of interest. The research can be undertaken in one of the Materials Science and Engineering research groups, in another research group of Delft University of Technology, outside the university in an Industrial environment, or in collaboration with another educational institute. The thesis has the same emphasis on original, independent research as a PhD thesis, but is of reduced scope. The Master Thesis Project culminates in a presentation and oral defence of the written thesis. Some examples of research projects are: - Welding aluminium alloys without causing residual stresses or distortion - Development of self-healing coatings for turbine blades - Recycling neodymium-magnet scrap - Computer simulation of friction and adhesion - Hydrogen storage for energy applications - Authenticity of Van Gogh paintings - Towards new light-emitting polymers - Micromechanics, fracture mechanics and chemomechanics of concrete - Thermomechanical fatigue of SiMo cast iron used in truck engines - Sustainability of building materials and environments
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MS53010-20	MSE Literature Research	10
Responsible Instructor	Y. Gonzalez Garcia	
Assistant	S.M. van der Meer	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	None (Self Study)	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The purpose of this course is to make a critical assessment of the literature covering a selected aspect of Materials Science and Engineering. The study is undertaken in preparation for the Master's graduation project (MS53035).	
Study Goals	The learning goals of the literature study are: - systematically collect and interpret extensive and relevant quantities of information from different perspectives and sources (industrial, societal and scientific); - summarize the information in a systematic overview representing the State-of-the-Art literature in the field of your MSc thesis and formulate a problem statement/research goal for your MSc Thesis project, if it applies; - take a systematic approach when performing a literature study - perform a thorough gap analysis - If the topic is related to your MSc thesis, identify the most relevant research questions and opportunities in the field of your thesis project Transferable skills you should demonstrate: - a scientific attitude (i.e. independent, critical and reflective); - the ability to provide a sound line of reasoning and to take a substantiated position; - the ability to draw conclusions that point towards scientific opportunities and to formulate research objectives; - a writing style and quality that equals the required level of the MSc thesis (follows the hourglass method) - a responsible and proactive attitude	
Education Method	Individual work. Self-study	
Prerequisites	The student must have approval from the MSE master coordinator BEFORE starting the literature review.	
Assessment	The deliverables of the literature review are: - a written report - a presentation both, to be completed before starting the graduation research project (MS53035). The assessment consists of - Written report to be graded - A 6 is the minimum grade to pass. - Presentation - Pass/Fail Criteria As part of the master thesis procedure, the student must attend and participate in the Q&A of at least ten (10) MSE colloquium. This should be completed before the graduation date	
Department	3mE Department Materials Science & Engineering	
Contact	You can contact the MSE master coordinator, Dr. Yaiza Gonzalez-Garcia by email: MSEcoordinator-3mE@tudelft.nl	

MS53035	MSE MSc Thesis	35
Responsible Instructor	Y. Gonzalez Garcia	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4 Different, to be announced	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The Master Thesis Project is the final program component, during which the student performs a literature survey (MS53010-20) and an independent scientific investigation that leads to a thesis (MS53035). This thesis may relate to the specialisation chosen or another specific area of interest. The research project can be undertaken in one of the Materials Science and Engineering research teams, in another research group of Delft University of Technology, outside the university in an Industrial environment, or in collaboration with another educational institute.	
Course Contents Continuation	Some examples of project topics: - From sand to grade silicon for solar energy application - Design of new materials with unprecedented properties by using machine learning - Design of 2D-materials for catalysis application - Study of 3D printing of materials: from microstructure to properties and performance - Bonding and corrosion failure in electronic devices and solar cells - Study of environmental-friendly corrosion inhibitors for the protection of metals - Influence of heat-treatment and alloying elements on the microstructure development of steels - Critical raw materials in the city: Recycling perspectives - Control of the mechanical properties of additively manufactured and functionally graded metal alloys	
Study Goals	The student is able to meet the requirements set in the exit qualifications of the master's program Materials Science and Engineering. More specifically, the student achieves the following qualifications: 1. A broad and profound knowledge of Materials Science and Engineering and the underlying engineering sciences (mathematics, physics and chemistry). 2. The capability to apply this knowledge at an advanced level at different levels of abstraction, including a reflective understanding of its structure and relations to other fields. 3. The capability for innovative contributions to the discipline in the form of new knowledge on materials or the development of new materials. 4. The capability to evaluate and assess the technological, ethical and societal impact of ones work, and to take responsibility with regard to sustainability, economy and social welfare. 5. The capability to solve technological problems in a systematic way, involving problem analysis, formulating sub-problems and providing innovative technical solutions, also in new and unfamiliar situations. 6. Professional attitude towards identifying and acquiring lacking expertise, monitoring and critically evaluating existing knowledge, planning and executing research. 7. The capability to work both independently and in multidisciplinary teams. Effective communication with both professionals and a non-specialised public.	
Education Method	Individual work. Self-study Research project	
Prerequisites	Before starting the Master Thesis Project (MS53010-20 & MS53035), the student should make sure that one of the staff members of the department Materials Science and Engineering takes responsibility for the project and subsequently acts as an MSE thesis supervisor. The student MUST obtain the green light from the master MSE coordinator BEFORE starting the literature review (MS53010-20), to comply with the following pre-requisite: has no more than 10ECs left in courses (at the start of the literature review) The student can start the research project (MS53035) when has completed the Literature Review (MS53010-20) or it is close to being finished (approx. 85%), only upon agreement with the MSE supervisor.	
Assessment	The individual student prepares a master's thesis to report on their project. The project work is then concluded through: - a public colloquium by the candidate (max. 30 min) - a defense in front of the MSc graduation committee (closed oral examination) The discussion (defense), presentation, report and performance will be assessed following the corresponding grading matrix (https://wbmt2.tudelft.nl/organisatie/formulieren/msc/3me-msc-masterthesis-rubrics.pdf) As part of the master thesis procedure, it is mandatory for the student to attend and participate in the Q&A of at least ten (10) MSE colloquium. This should be completed before the graduation date	
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